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CR-MSNet: a dual-branch multi-scale attention network for multi-label chest X-ray classification

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Multi-label disease diagnosis in chest X-rays necessitates simultaneous consideration of both global organ structures and local lesion characteristics. However, current methodologies primarily utilize single-branch architectures and lack effective attention guidance mechanisms, which complicates the balance between global context and local details. Furthermore, multi-label datasets for chest X-rays often suffer from significant class imbalance. We propose CR-MSNet, a dual-branch multi-scale attention network designed for multi-label chest X-ray classification. The global branch is constructed using CoAtNet-2-rw to capture holistic semantic representations, while the local branch employs a residual convolutional neural network to extract detailed lesion features. We incorporate a cross-attention mechanism to facilitate adaptive interaction and information exchange between global and local representations. Additionally, we propose a Parallel Multi-Scale Channel-Spatial Attention (PMS-CSA) module to enhance both key semantic channels and potential lesion regions, thereby increasing the discriminative power of feature representations. A two-stage training strategy with an adjusted loss function is implemented to effectively alleviate the detrimental effects of class imbalance on model performance. Experimental results indicate that CR-MSNet achieves a macro-average AUC of 0.847 on the ChestX-ray14 dataset, confirming its effectiveness and potential for application in multi-label classification tasks for chest X-rays. By seamlessly integrating a dual-branch architecture with multi-scale attention mechanisms, this study confirms the critical role of attention-guided feature interactions in reconciling global and local representations.

Keywords Multi-label classification, Dual-branch network, Attention mechanism, Data imbalance, Chest X-ray

Chest diseases represent a persistent threat to public health, characterized by diverse clinical phenotypes and high incidence rates. Timely and accurate diagnosis is essential for enhancing patient prognosis. Chest X-Ray (CXR) has emerged as the most widely utilized imaging modality in clinical chest examinations, owing to its rapid imaging capabilities, low radiation dose, and cost-effectiveness. However, the interpretation of traditional CXR relies heavily on the clinical experience and expertise of radiologists, rendering the diagnostic process both time-consuming and highly subjective¹. Consequently, the development of high-precision, automated computer-aided diagnostic methods for chest diseases holds significant clinical value. CXR multi-label classification aims to identify multiple chest diseases from a single image; however, the task remains exceedingly challenging due to the complex anatomical structure of the chest, substantial variability in lesion size, and extreme class imbalance within the dataset.

In recent years, deep learning technology has significantly advanced the field of CXR multi-label classification. CheXNet² was the first to demonstrate the potential of convolutional neural networks (CNNs) for this task. Subsequently, numerous studies have focused on enhancing performance by refining network architectures, incorporating attention mechanisms, and developing multi-scale feature modeling strategies. For instance, Chen et al.³ designed a dual-path convolutional structure based on DenseNet121 and implemented a scale-aware attention module to more effectively capture multi-scale features. Similarly, Chowdary et al.⁴ constructed a framework utilizing a dual-branch network and employed two pre-trained AlexNet models to extract discriminative features, thereby improving classification performance. However, despite the demonstrated advantages of dual-branch structures in enhancing feature representation, most existing hybrid architectures continue to rely on CNNs as the backbone network. This reliance results in feature modeling that is biased

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toward local textures, making it challenging to fully capture long-range dependencies and global semantic relationships in chest images. Furthermore, the CNN-dominated architectures are susceptible to interference from large structural contrast regions, such as heart shadows and ribs, which limits their effectiveness in fine-grained anomaly recognition and leads to insufficient attention to small lesions. In summary, multi-label classification of CXRs still faces three key challenges:

- (1) Multi-scale lesions present significant challenges for uniform modeling. CXRs exhibit complex multi-scale abnormalities, ranging from small nodules to extensive volumes of effusion, making it difficult for a single structure to simultaneously capture both global information and local fine features.
- (2) Current attention mechanisms often fail to accommodate the intricate anatomical structures present in chest radiographs. Traditional attention modules typically conduct channel or spatial modeling using a single receptive field, which hinders their ability to effectively capture multi-scale lesion information. Furthermore, the attention generation process is prone to interference from larger structures, such as cardiac shadows and ribs, resulting in an inadequate response to small lesion features.
- (3) Extreme data imbalance results in inadequate learning of minority classes. In the ChestX-ray14 dataset, certain diseases are represented by a limited number of samples, which causes models to prioritize training on majority classes. This bias ultimately leads to diminished predictive performance for minority classes.

This study introduces a dual-branch multi-scale attention network, CR-MSNet, which employs CoAtNet-2-rw⁵ as the global Transformer backbone and utilizes residual convolutional neural networks to form local branches. This parallel architecture facilitates the integration of global semantics with local fine-grained features, thereby significantly enhancing overall recognition capabilities. Furthermore, we designed a parallel multi-scale channel-spatial attention mechanism (PMS-CSA) to improve the model's focus on critical lesion areas, resulting in more accurate detection of small lesions and minority classes. To mitigate the detrimental effects of extreme data imbalance on feature learning for underrepresented classes, this study implements a two-stage training strategy with an adjusted loss function to increase the model's sensitivity to low-frequency classes. Experimental results demonstrate that the proposed dual-branch structure surpasses single-branch multi-label classification performance. The introduction of the parallel multi-scale channel-spatial attention mechanism (PMS-CSA) in both branches further enhances the model's feature representation capabilities. When combined with a cross-attention fusion strategy, the model achieves optimal performance, thereby validating the effectiveness and potential for application of the proposed method.

Related work

Multi-label CXR classification

Multi-label classification of CXR presents greater challenges than single-label tasks due to the potential for multiple abnormalities within a single image. Early studies predominantly utilized CNNs, achieving notable advancements by capitalizing on their local inductive bias and region-sensing capabilities. Wang et al.⁶ conducted a systematic evaluation of various classical CNN architectures, including AlexNet, GoogLeNet, VGG-16, and ResNet-50, on the ChestX-ray14 dataset. Strick et al.⁷ and Zhao et al.⁸ further improved the models' receptive fields and contextual modeling abilities through structural optimization and large-kernel convolution mechanisms, respectively. In addition to CNNs, Transformers have also exhibited robust global modeling capabilities. Taslimi et al.⁹ developed a multi-label classification model based on the Swin Transformer, utilizing a multilayer perceptron as a prediction head to enhance disease representation. Öztürk et al.¹⁰ explicitly modeled disease co-occurrence relationships through multi-branch weighted outputs and cross-label aggregation mechanisms, resulting in improved classification performance. As research progressed, hybrid architectures that integrate CNNs and Transformers gained prominence. Ashraf et al.¹¹ conducted a comparison of multi-class models on ChestX-ray14, confirming the superior performance of CoAtNet. Li et al.¹ advanced this field by proposing the CheX-DS model, which integrates DenseNet and the Swin Transformer through model ensemble. Similarly, Faisal et al.¹² combined the Vision Transformer (ViT) architecture with CheXNet. Additionally, Lee et al.¹³ developed a hybrid architecture, CheXGAT, which merges convolutional neural networks with graph neural networks, thereby improving classification performance by modeling potential associations among diseases. Ngo et al.¹⁴ proposed the Dual Dynamic Consistency Regularization (D²CR) framework, which introduces dynamic consistency regularization and a threshold-based filtering mechanism within a dual-branch architecture to improve pseudo-label quality and enhance cross-domain generalization. Subsequently, they developed the Explicitly Class-specific Boundaries (ECB)¹⁵ hybrid model, which integrates the global modeling capability of Vision Transformers (ViT) with the local feature learning strength of CNNs to achieve finer target-domain modeling. Through collaborative training, high-quality pseudo labels are generated to reduce inter-branch knowledge discrepancies, thereby improving generalization performance. In addition, Bui et al.¹⁶ proposed the CLEAR framework, which combines global and local Transformer structures, incorporates a dual-branch cross-fusion mechanism and channel-aware self-attention (CASA), and further aligns semantic representations using a pretrained language model to enhance feature representation and high-level semantic modeling capacity.

Existing methods, however, exhibit several limitations. CNNs emphasize local texture modeling, whereas Transformers, despite their ability to model global features, demonstrate limited sensitivity to fine-grained small lesions. Hybrid architectures primarily utilize serial or post-fusion strategies, which do not fully realize unified modeling of multi-scale lesions or effectively tackle minority class recognition. To overcome these challenges, this paper introduces a dual-branch parallel architecture that integrates attention mechanisms with a two-stage training strategy. This innovative approach facilitates the collaborative modeling of global and local features, thereby enhancing detection capabilities for small lesions and minority categories.

Multi-scale feature extraction

Multi-scale feature extraction plays a vital role in CXR analysis due to the significant variation of chest abnormalities across different scales. To effectively model lesions at these varying scales, numerous approaches have been proposed in the literature. For example, Xiong et al.¹⁷ introduced the CONVFCMAE framework, which facilitates multi-scale feature extraction by partially freezing the backbone network and integrating global pooling with multi-head attention. Similarly, Yu et al.¹⁸ developed CD-DETR, which utilizes multi-scale features in conjunction with channel and spatial attention to optimize the detection performance of small lesions. Wang et al.¹⁹ implemented a deep fusion of multi-scale features through recursive segmentation and semantic interaction mechanisms, whereas Alqahtani et al.²⁰ utilized progressive downsampling and ViT self-attention for the adaptive aggregation of multi-scale features. Building upon these methodologies, we present a dual-branch architecture: the local branch emphasizes fine-grained features of small lesions using ResNet18²¹, while the global branch captures overall structural information through CoAtNet-2-rw. Attention modules are integrated at critical stages, facilitating global–local feature fusion via Cross-Attention to enhance the expression capabilities for multi-scale lesions.

Attention mechanisms

Attention mechanisms improve feature representation and discrimination by adjusting feature weights to emphasize critical regions. Squeeze-and-Excitation (SE)²² module identifies significant features through global semantic compression and channel re-calibration. Convolutional Block Attention Module (CBAM)²³ utilizes sequential attention across both channels and spatial dimensions to optimize feature responses progressively. Efficient Channel Attention (ECA)²⁴ enhances expressiveness at a low computational cost by modeling cross-channel dependencies using one-dimensional convolutions without reducing dimensionality. These attention mechanisms are extensively employed in multi-label classification tasks for CXRs. Singh et al.²⁵ integrated multi-head self-attention into DenseNet121 to enhance feature modeling. Pan et al.²⁶ introduced sliding-window attention to increase local modeling efficiency. Xu et al.²⁷ developed a synergistic structure that combines image-level and fine-grained disease attention to enhance lesion focus. Ucan et al.²⁸ incorporated coordinate attention into EfficientNetB7 to boost multi-class disease recognition performance. Jiang et al.²⁹ constructed a Transformer-based dual-path attention decoder to improve the modeling of associations between local lesions and labels. In addition, Wan et al.³⁰ proposed the Multi-order Cross Geometry-aware (MCG) model, which captures long-range dependencies between spatial locations through higher-order correlation modeling and integrates cross-layer attention to achieve multi-scale geometric representation. Building upon this foundation, they further developed the Implicit Multi-order Correlating Geometry-aware (IMCG)³¹ model, which introduces multi-order correlation modeling in both spatial and channel dimensions to enable more refined feature recalibration and structural representation. Subsequently, they proposed the Multi-Head Ranking Attention (MHRA)³² mechanism, which distinguishes between highly correlated and weakly correlated features for hierarchical fusion, thereby alleviating the tendency of conventional self-attention to focus excessively on salient regions and enhancing the modeling of fine-grained structural information.

Although current attention mechanisms have enhanced multi-label classification performance, most architectures continue to depend on single receptive fields or serial attention methods. This reliance limits their adaptability to multi-scale structures and inadequately addresses minority classes and small lesions. To overcome these limitations, this study introduces the Parallel Multi-Scale Channel-Spatial Attention (PMS-CSA) module. By significantly enhancing feature expression capabilities across three dimensions—channel attention module, spatial attention module, and fusion module—it effectively improves the model's capacity to identify complex-scale lesions and long-tail categories, thereby substantially increasing the diagnostic reliability of CXR multi-label classification.

Data imbalance

The multi-label classification task in CXR often encounters the challenge of data imbalance. Deep networks typically prioritize learning classes with larger sample sizes, while those with fewer samples are frequently underrepresented in the training process. To mitigate this issue, Albahli et al.³³ improved the visibility of minority classes through synthetic data augmentation. Nguyen-Mau et al.³⁴ integrated Mixup, CutMix, Mosaic, and Focal Loss with high-resolution input augmentation to enhance fine-grained recognition capabilities. Cai et al.³⁵ approached the imbalance by optimizing feature distances using adaptive weights and label clustering. Hanif et al.³⁶ introduced FZLPR loss to concentrate on hard-to-classify samples. Seo et al.³⁷ improved modeling of tail categories through class-balanced sampling and multi-expert structures. While these methods effectively address class imbalance through data augmentation, loss reconstruction, and sampling strategies, most utilize single-stage training. This paper proposes a two-stage training strategy to overcome these limitations.

Dataset and training settings

Dataset

ChestX-ray14 is a large-scale public dataset of CXRs released by the National Institutes of Health (NIH). It ranks among the most widely utilized multi-label CXR classification benchmark datasets in the field of medical imaging. The dataset comprises 112,120 anteroposterior CXR images sourced from 30,805 independent patients. All labels were derived from the corresponding medical reports through a natural language processing model, encompassing 14 disease classes: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural_Thickening, and Hernia. The pathological distribution of the dataset is illustrated in Fig. 1.

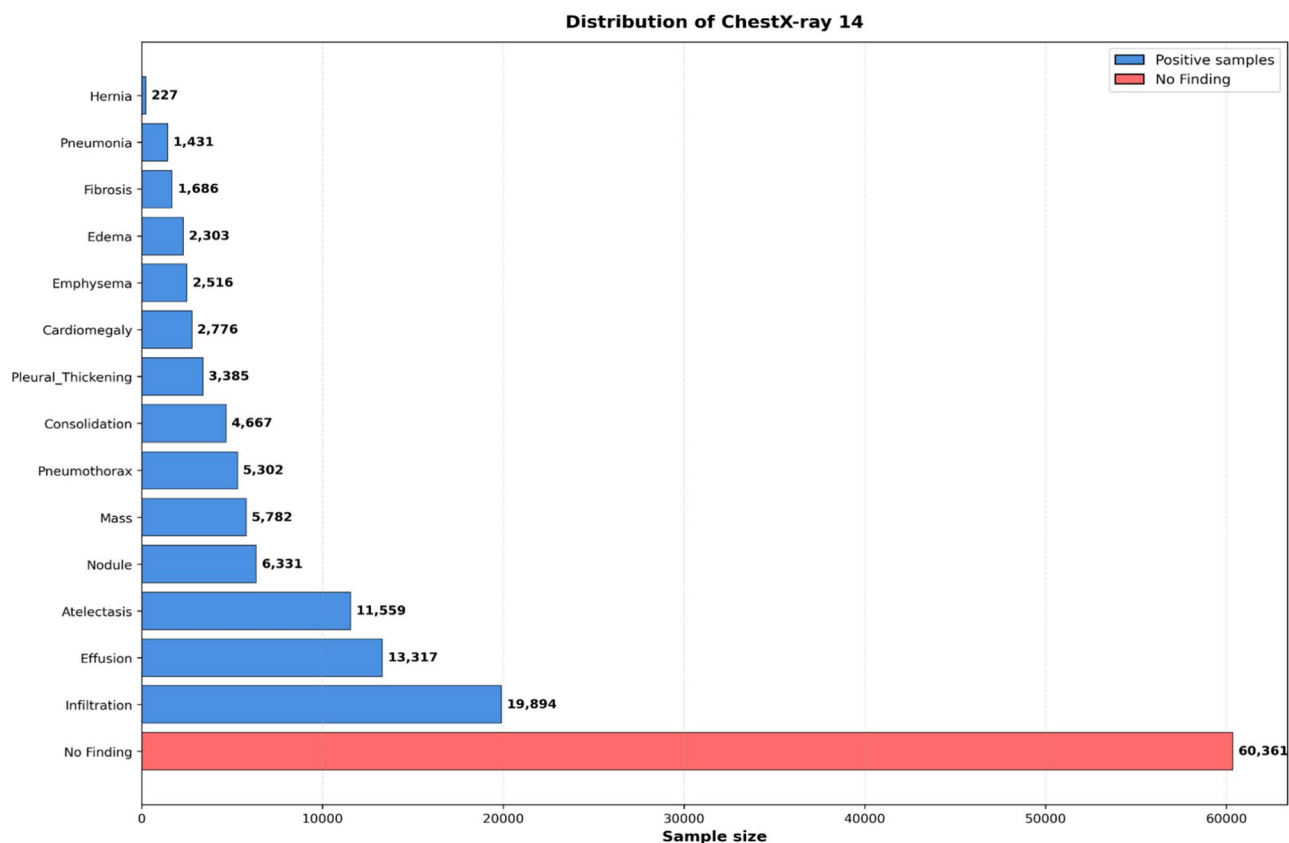


Fig. 1. Pathology distribution in ChestX-ray14 dataset.

Data partitioning and preprocessing

This study utilized a patient-level stratified random partitioning strategy, creating training, validation, and test sets based on patient IDs in a 7:1:2 ratio. This approach ensured that all images from the same patient were allocated to the same partition and that the prevalence of each disease was consistent across all three partitions, thereby preventing data leakage and enhancing model generalization. To assess the validity of this partitioning, we statistically analyzed the number and proportion of positive results for 14 diseases across the training, validation, and test sets (Figs. 2, 3). The findings indicated that the distribution of various diseases remained uniform across the different subsets without significant bias. Additionally, to further investigate whether the statistical relationships among diseases in the multi-label task were consistent across subsets, we constructed co-occurrence rate matrices for each subset (Figs. 4, 5 and 6). The visualization results demonstrated that the co-occurrence rate matrices across subsets exhibited a high degree of consistency in both overall structure and local hotspot regions, suggesting that the subsets effectively preserved the true distribution characteristics of multiple co-occurring diseases in ChestX-ray14. These results confirm that the data partitioning in this study was equitable and stable, maintaining the original statistical relationships between diseases and providing a robust foundation for model training and evaluation.

The ChestX-ray14 dataset comprises single-channel grayscale images. To align with the three-channel input structure of the CoAtNet-2-rw pretrained model, this study replicates the grayscale images into three channels prior to inputting them into the network, thereby ensuring consistency with the pretrained weight structure. The original image resolution is 1024×1024 . To enhance computational efficiency and adhere to the input specifications of the backbone network, all images are uniformly resized to 224×224 . Furthermore, all input images are normalized using the mean and standard deviation of ImageNet. For data augmentation, this study employs a combined phased augmentation strategy. Spatial augmentations include RandomResizedCrop (minimum crop ratio 0.8), random horizontal flipping ($p=0.5$), random rotation ($\pm 15^\circ$, $p=0.5$), and random scaling (up to $1.2 \times$, $p=0.3$) to simulate natural variations in lesion location, perspective, and scale. Given that this task involves multi-label disease presence classification without lateral localization prediction, horizontal flipping does not alter label semantics while reducing the risk of model overfitting to fixed anatomical orientation priors. Rotation angles are restricted to $\pm 15^\circ$ to reflect minor positioning deviations in clinical imaging and to prevent unreasonable anatomical distortion. Additionally, brightness and contrast perturbations ($p=0.3$) are introduced to accommodate variations in grayscale distributions across different imaging conditions. All augmentation operations were conducted at the batch level. Similar data augmentation strategies^{2,11,38} have been employed in multiple studies utilizing the ChestX-ray14 dataset.

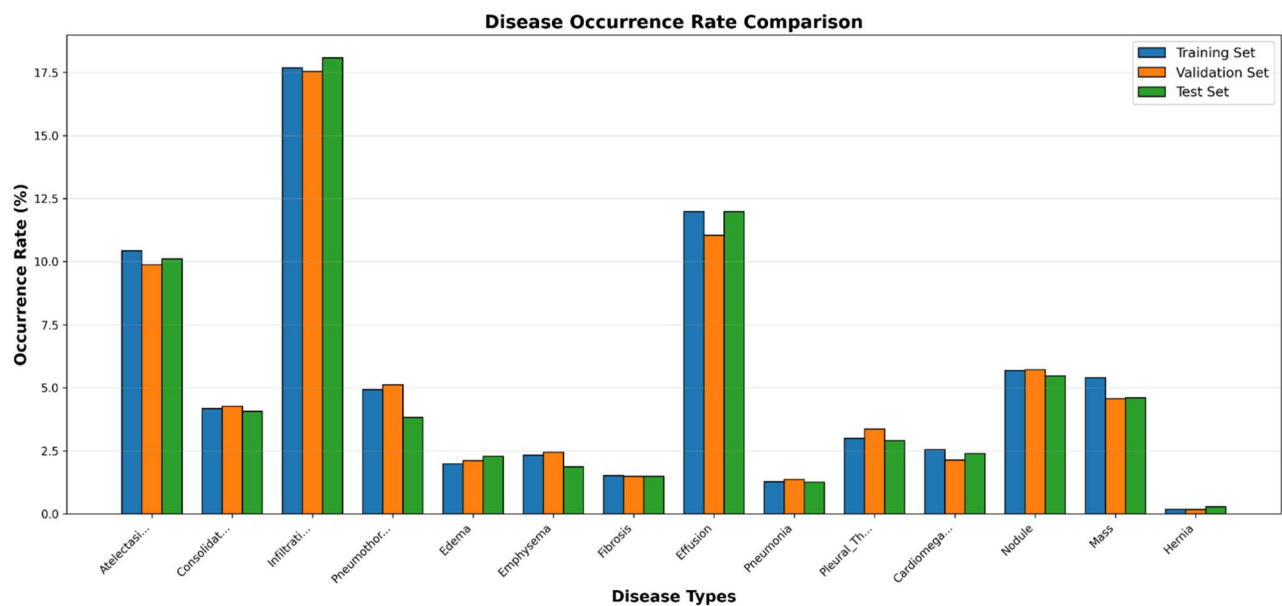


Fig. 2. Proportions of positive labels for the 14 diseases in the training set, validation set, and test set. There is no significant deviation in the proportion of each disease across the three partitioned datasets.

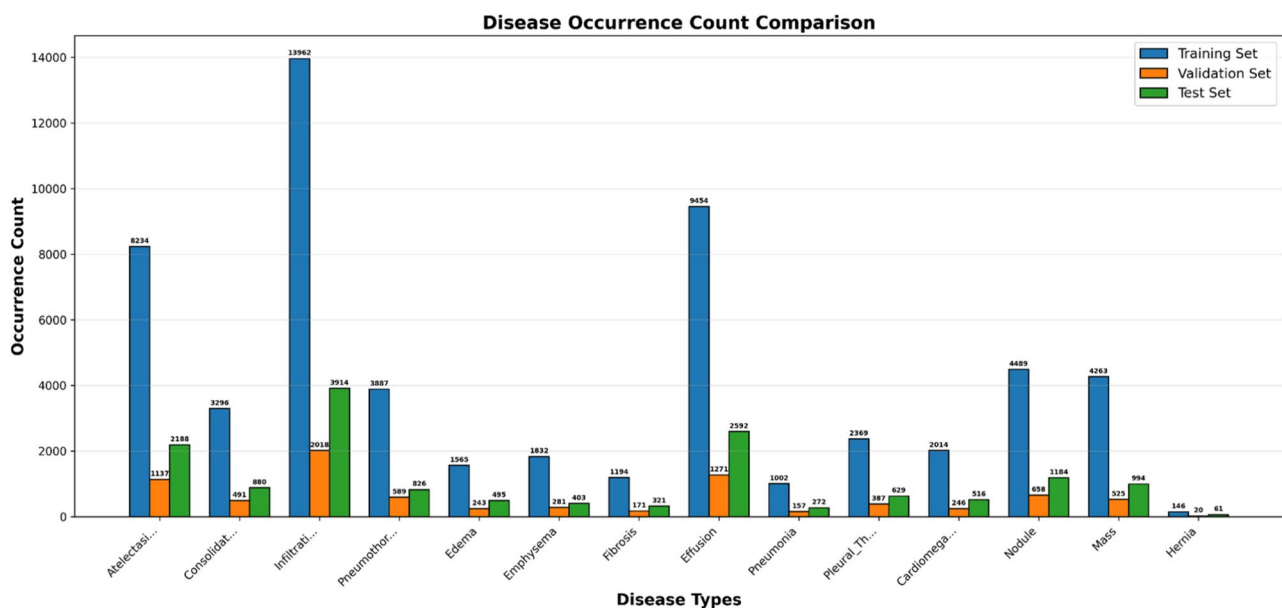


Fig. 3. Absolute distributions of positive labels for each disease across the training set, validation set, and test set.

Training settings

This study utilizes the PyTorch deep learning framework for model construction and training. All experiments were conducted on a server equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM), a 16 vCPU Intel(R) Xeon(R) Platinum 8352 V CPU @ 2.10 GHz, and 120 GB of system memory. The operating system was Ubuntu 20.04, with Python version 3.8, PyTorch version 1.11.0, and CUDA version 11.3. A two-stage optimization strategy was implemented during training. The AdamW optimizer was chosen, with a batch size of 32 and a fixed random seed of 85. The weight decay coefficients for the first and second stages were set at 0.001 and 0.01, respectively. The first stage involved 15 epochs of training. To stabilize gradient updates during the initial fine-tuning phase of the pre-trained model, a linear warmup strategy was employed, gradually increasing the learning rate from near 0 to 1×10^{-3} over the first 2 epochs. Subsequently, the first stage of training is completed with the learning rate held constant. Stage 2 continued training on the stage 1 model, reducing the learning

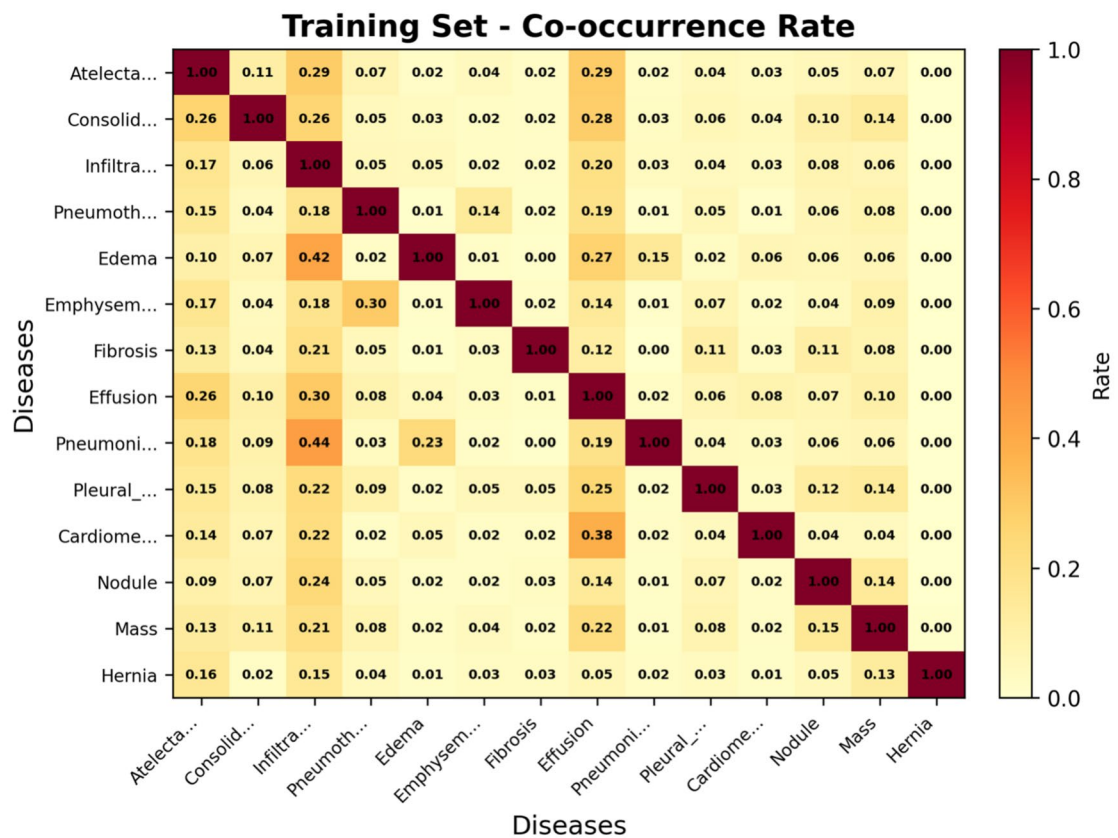


Fig. 4. Disease co-occurrence rate matrix for the training set.

rate to 1×10^{-4} and training for 25 epochs to achieve fine-grained model refinement. To mitigate overfitting, an early stopping mechanism was implemented; training was terminated prematurely when the validation set loss failed to improve over 8 consecutive epochs. Both branches are initialized using ImageNet pre-trained weights. To ensure reproducibility, the random seed was fixed. Three independent runs yielded consistent performance, with an mAUC of 0.847 in each run.

Methods

Overall architecture

We propose CR-MSNet, a dual-branch multi-scale attention network designed for the multi-label classification of CXRs. The overall architecture of the model is illustrated in Fig. 7. The global branch is constructed using CoAtNet-2-rw, while the local branch follows a ResNet-18-style architecture. The PMS-CSA module is introduced at the high-level semantic stage of the global branch and at the feature enhancement stage of the local branch. A cross-attention mechanism facilitates the complementary fusion of global and local features. Combined with a two-stage training strategy to optimize feature learning, the model ultimately enables multi-label prediction for 14 chest diseases.

Dual-branch architecture

We select CoAtNet-2-rw as the global branch network for global semantic modeling. Its architecture adheres to a progressive design concept of “pre-convolution, post-attention,” which integrates the benefits of convolutional operators and self-attention mechanisms at various stages of network depth. The overall network comprises five stages, including two convolutional modules, eight MBCConv modules, and sixteen relative self-attention modules. The initial stage employs convolutional layers to transform the input image from the spatial domain to a high-dimensional feature domain, facilitating the initial encoding of fundamental local structures. Subsequently, Stages 1 and 2 consist of lightweight convolutional modules based on MBCConv. By utilizing depthwise separable convolution, these stages effectively concentrate on local lesion regions and their surrounding spatial context while maintaining computational efficiency, thereby enhancing the representation of fine-grained textures and anatomical structures. Building on this foundation, Stages 3 and 4 introduce Transformer-based relative self-attention modules. Through a global self-attention mechanism, these modules model long-range dependencies among cross-regional features, capturing potential lesion association patterns distributed across different spatial locations. By leveraging the strengths of convolutional structures for local feature extraction and the capabilities of Transformers for global dependency modeling, CoAtNet-2-rw progressively constructs stable and discriminative global semantic representations at deeper levels, thus playing a pivotal role in global feature

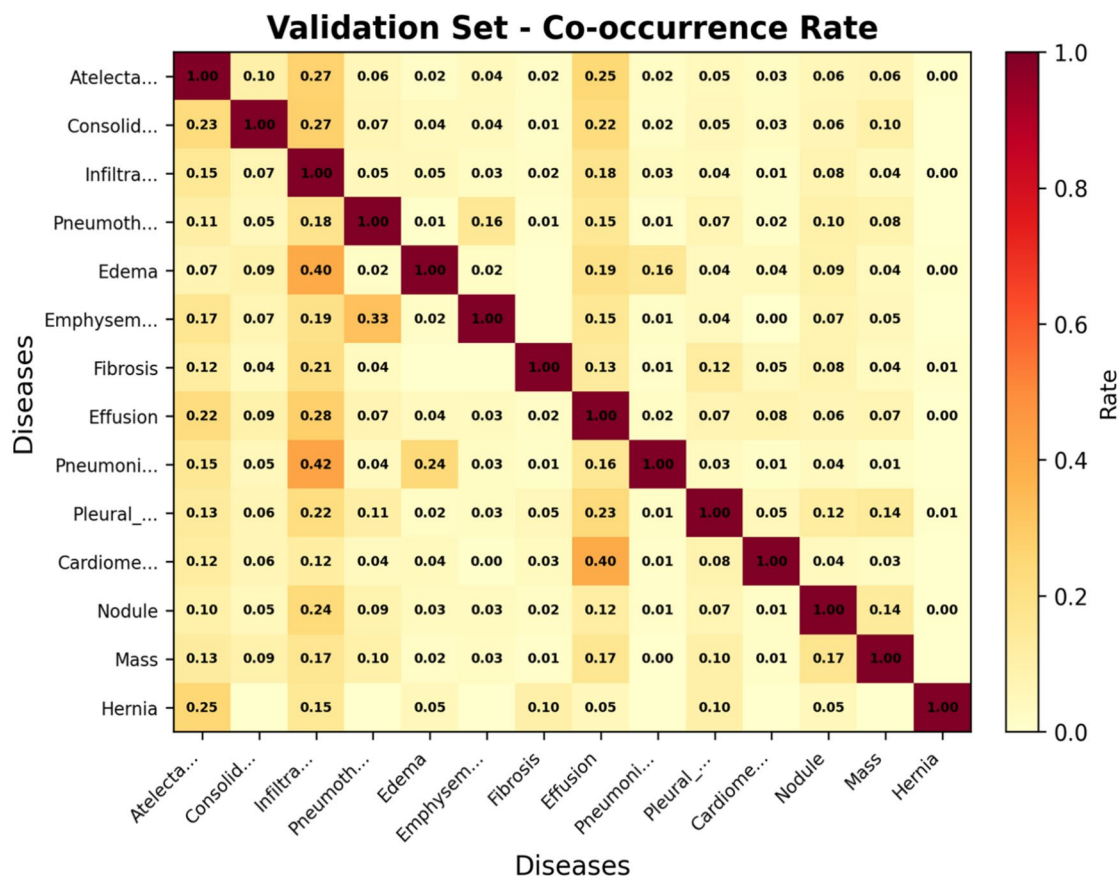


Fig. 5. Disease co-occurrence rate matrix for the valid set.

modeling within the overall framework. During feature evolution, the input image is initially downsampled to a uniform resolution of 224×224 . It then proceeds through five stages of feature extraction and scaling, which progressively reduce the spatial dimensions from 224×224 to 112×112 , 56×56 , 28×28 , 14×14 , and finally to 7×7 . Concurrently, the number of channels increases from 3 to 64, 128, 256, 512, and ultimately to 1024. This strategy of progressive downsampling and channel expansion effectively compresses spatial redundancy while enhancing the semantic abstraction of features, thereby yielding high-quality global feature representations for subsequent high-level semantic modeling and attention enhancement.

The local branch utilizes a convolutional neural network architecture based on ResNet18, which is specifically designed to model local anatomical structures and fine-grained lesion patterns in chest X-ray images. Upon receiving the input image, this branch first processes it through a 7×7 convolutional layer with a stride of 2, followed by batch normalization and a ReLU activation function. This sequence enhances the receptive field while extracting low-level features, such as edges and textures, and reduces the spatial resolution from 224×224 to 112×112 . Subsequently, a 3×3 max pooling layer with a stride of 2 further downsamples the image to 56×56 , thereby decreasing computational complexity while preserving essential structural information. Following the initial encoding, the network sequentially stacks four progressively increasing residual blocks (layer 1–layer 4), with channel numbers set to 64, 128, 256, and 512, respectively. In this configuration, layer 1 conducts preliminary modeling of local features while preserving spatial resolution. Layers 2 through 4 incorporate convolutional operations with a stride of 2 in their first residual unit, thereby compressing the feature map dimensions to 28×28 , 14×14 , and 7×7 . This strategy of progressive downsampling and channel expansion reduces spatial redundancy while enhancing the semantic abstraction of features, effectively capturing high-resolution local details and structural lesion information. Upon completion of multi-layer residual feature extraction, a feature enhancement module, comprising two 3×3 convolutional layers, is introduced to further expand the channel dimension and augment feature expressiveness while maintaining spatial resolution, mapping local semantics to a unified 1024-dimensional feature space. Subsequently, the feature map is standardized to a 7×7 scale using AdaptiveAvgPool2d to ensure consistency with the global branch output features regarding spatial dimensions, thereby establishing a structural foundation for subsequent dual-branch feature fusion.

Formal mathematical description of the dual-branch

Let the initial input image tensor be denoted as $X \in \mathbb{R}^{B \times C \times H \times W}$, where B denotes the batch size, C denotes the number of channels, and H and W correspond to the spatial height and width, respectively. In the global branch, the input feature first undergoes an initial 3×3 convolution:

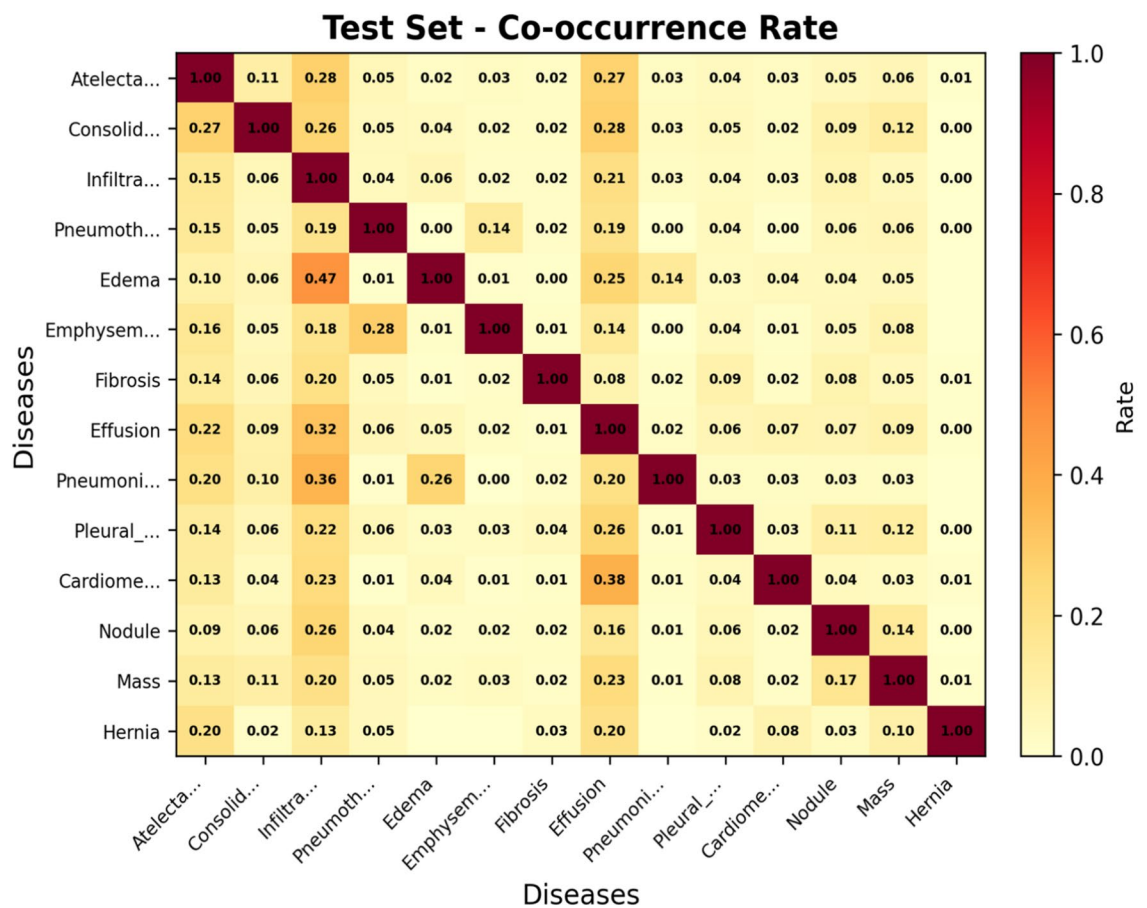


Fig. 6. Disease co-occurrence rate matrix for the test set.

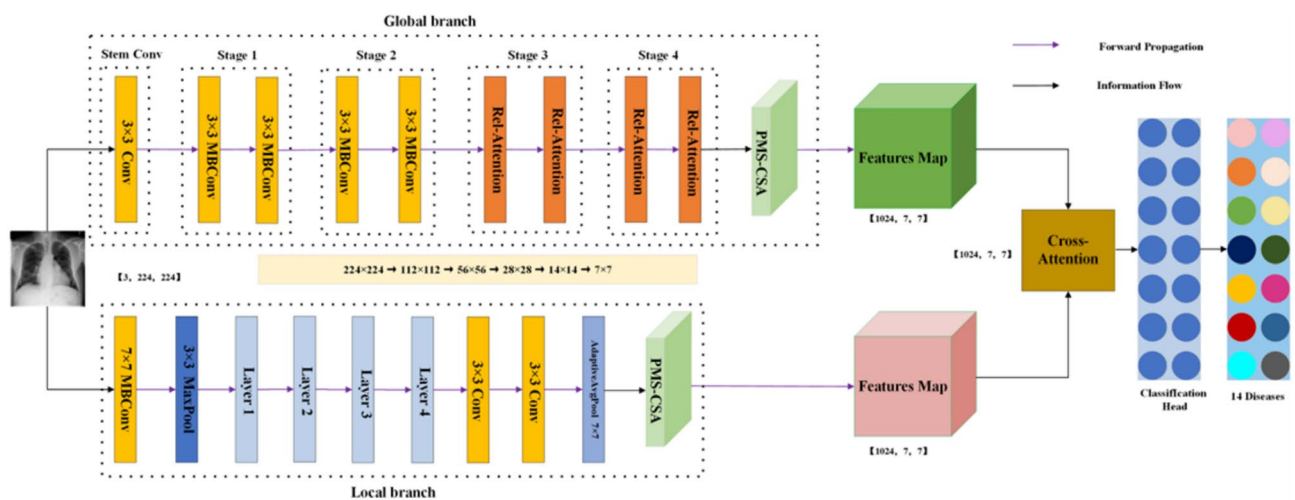


Fig. 7. Model Architecture Diagram. The global branch adopts CoAtNet-2-rw to integrate convolutional and Transformer-based representations, while the local branch is built on ResNet18 for fine-grained feature extraction. The PMS-CSA module is embedded at the high-level semantic stage of the global branch and at the feature enhancement stage of the local branch. Features from both branches are fused via a cross-attention module and fed into a classification head to predict 14 disease labels.

$$\mathbf{G}_0 = \text{Conv}_{3 \times 3}(\mathbf{X}) \quad (1)$$

where $\mathbf{G}_0$ denotes low-level features that preserve spatial structure. Subsequently, multiple MBConv modules are stacked to progressively enlarge the receptive field:

$$\mathbf{G}_1 = \text{MBConv}_1(\mathbf{G}_0), \mathbf{G}_2 = \text{MBConv}_2(\mathbf{G}_1) \quad (2)$$

where $\mathbf{G}_1$ captures local contextual features and $\mathbf{G}_2$ encodes mid-scale anatomical semantic representations. To further model long-range dependencies in higher-level feature space, Transformer modules are introduced:

$$\mathbf{G}_3 = \text{Transformer}_1(\mathbf{G}_2), \mathbf{G} = \text{Transformer}_2(\mathbf{G}_3) \in \mathbb{R}^{B \times C_g \times H_g \times W_g} \quad (3)$$

where $\mathbf{G}_3$ denotes globally relational-aware features, and $\mathbf{G}$ denotes the high-level global semantic feature map of the global branch before applying the PMS-CSA attention module. For the local branch, the input image first passes through a 7×7 convolutional layer for initial feature extraction, followed by Batch Normalization (BN) and a ReLU activation function for feature normalization and nonlinear transformation. A max-pooling operation is then applied for downsampling:

$$\mathbf{L}_0 = \text{MaxPool}(\delta(\text{BN}(\text{Conv}_{7 \times 7}(\mathbf{X})))) \quad (4)$$

where $\delta(\cdot)$ denotes the ReLU activation function. Next, four residual stages are employed for hierarchical modeling:

$$\mathbf{L}_1 = \text{Layer}_1(\mathbf{L}_0), \mathbf{L}_2 = \text{Layer}_2(\mathbf{L}_1), \mathbf{L}_3 = \text{Layer}_3(\mathbf{L}_2), \mathbf{L}_4 = \text{Layer}_4(\mathbf{L}_3) \quad (5)$$

where $\mathbf{L}_1$ represents shallow local texture features, $\mathbf{L}_2$ encodes mid-level structural features, $\mathbf{L}_3$ captures deep lesion-responsive representations, and $\mathbf{L}_4$ denotes highly discriminative semantic features. To further enhance the discriminative capability of the local branch, a feature enhancement convolutional module is introduced after $\mathbf{L}_4$. This module consists of two 3×3 convolution layers with BN and ReLU inserted between them, forming a typical Conv-BN-ReLU-Conv structure:

$$\mathbf{L}_{enh} = \text{Conv}_{3 \times 3}(\delta(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{L}_4)))) \quad (6)$$

An adaptive average pooling operation is then applied for spatial alignment:

$$\mathbf{L} = \text{AdaptiveAvgPool2D}(\mathbf{L}_{enh}) \in \mathbb{R}^{B \times C_l \times H_l \times W_l} \quad (7)$$

where $\mathbf{L}$ denotes the local branch feature map before enhancement by the PMS-CSA attention module.

Parallel multi-scale channel-spatial attention (PMS-CSA) module

Given the significant differences in lesion scale, the weak representation of local abnormalities, and the highly imbalanced class distribution in multi-label CXR classification tasks, a single backbone network faces challenges in achieving both global semantic consistency and local lesion discriminability at the feature modeling level. While CoAtNet-2-rw effectively captures long-range spatial dependencies at the high-level semantic stage through relative position self-attention, its Transformer backbone remains limited by insufficient inductive bias in local structure perception and fine-grained lesion localization. Consequently, this study integrates PMS-CSA into the high-semantic output layer of the global branch and the feature enhancement stage of the local branch to facilitate cross-scale feature compensation and enhance discriminative capabilities. In the global branch, PMS-CSA is implemented following Stage 4 of CoAtNet-2-rw. At this stage, the features exhibit the highest degree of semantic abstraction and the largest channel dimensions, having integrated ample global contextual information, which renders them appropriate as representations of disease-level discriminative features. The incorporation of channel and spatial attention at this juncture facilitates the adaptive recalibration of high-dimensional semantic channels, suppresses redundant or weakly correlated responses, and accentuates the global semantic patterns most pertinent to multi-label disease prediction. This approach enhances the model's discriminative stability and robustness in complex comorbidity scenarios. In the local branch, PMS-CSA is situated following the enhancement convolution module. This placement aligns with a crucial phase in the transition from low-level texture to lesion-level semantics within local features. At this juncture, the features maintain a distinct spatial geometric structure and local morphological information, rendering them optimal inputs for modeling fine-grained lesion patterns. By incorporating PMS-CSA at this stage, the model can selectively enhance channel features pertinent to local lesion discrimination and emphasize potential abnormal regions through spatial saliency modeling. This approach effectively addresses the limitations of the Transformer backbone in representing small-scale lesions and in achieving spatial localization. In the high-level semantic stage of the global branch and the feature enhancement stage of the local branch, the PMS-CSA module facilitates dual calibration of global semantic consistency and local discriminative features. This dual-branch attention deployment strategy mitigates redundancy and instability that may arise from excessive stacking of attention mechanisms within a single path. Additionally, it allows the model to dynamically balance the selection of disease-related features while preserving structural information across multi-scales and semantic levels, thereby enhancing the robustness of overall feature representation and generalization ability. The PMS-CSA module is

illustrated in Fig. 8, comprises three core components that collectively improve feature representation from three dimensions: Channel Attention module, Spatial Attention module, and Fusion module.

- (1) CA module: This module initially applies both global average pooling and global max pooling along the spatial dimension of the input features to obtain complementary channel-wise statistical descriptors. The resulting outputs are then summed, permuted, and input into a 1D convolution layer to model local dependencies among channels. The kernel size for this 1D convolution is set to 3, effectively capturing semantic correlations between adjacent channels while preserving a low parameter count. Subsequently, learnable per-channel affine modulation parameters are introduced to recalibrate the channel responses, and a Sigmoid activation function is employed to generate channel attention weights, thereby adaptively enhancing the discriminative capacity of semantic channels.
- (2) SA module: In parallel with channel attention, this module computes the mean and maximum values separately across the channel dimension to create a two-dimensional spatial saliency projection. These two maps are concatenated and subsequently processed through parallel dual-scale convolutional layers, with kernel sizes of 3×3 and 5×5 , respectively. The 3×3 convolution captures fine-grained local texture features, whereas the 5×5 convolution models spatial structural information within a larger receptive field. The outputs from the dual-scale convolution are fused using an equal-weight fusion strategy, and Dropout2d (0.1) is applied in the spatial dimension to enhance training stability and mitigate spatial overfitting. Ultimately, the Sigmoid activation function produces spatial attention weights that emphasize key discriminative regions.
- (3) Fusion module: To balance the contributions of channel-enhanced and spatial-enhanced features, the module incorporates two learnable scalar parameters that adjust the weights of the channel attention path and the spatial attention path, respectively. The fusion weights for both paths are initialized at 0.5 to ensure relatively balanced contributions from each attention type during the initial stages of training. Dynamic adjustment is subsequently facilitated through Sigmoid activation and normalization operations, allowing for adaptive weighted fusion. Additionally, an identity mapping is explicitly maintained as a third path to preserve original structural information and ensure gradient propagation stability. The final output is derived from the channel-augmented branch, the spatial-augmented branch, and the identity mapping path, thereby achieving a dynamic trade-off among semantic selection, spatial localization, and feature stability.

Formal mathematical description of the PMS-CSA module

Before enhancement by the PMS-CSA module, the feature maps of the global and local branches are denoted as $G \in \mathbb{R}^{B \times C_g \times H_g \times W_g}$, $L \in \mathbb{R}^{B \times C_l \times H_l \times W_l}$ respectively. Since both branches adopt the same attention mechanism structurally, the following description takes the global feature G as an example. The local branch L follows an identical procedure. To maintain structural stability during feature transformation, an explicit identity mapping G is preserved within the module. To obtain stable and complementary channel-wise statistical information, global average pooling and global max pooling are applied along the spatial dimensions:

$$\text{Avg}(G) = \text{AdaptiveAvgPool2d}(G) \in \mathbb{R}^{B \times C_g \times 1 \times 1} \quad (8)$$

$$\text{Max}(G) = \text{AdaptiveMaxPool2d}(G) \in \mathbb{R}^{B \times C_g \times 1 \times 1} \quad (9)$$

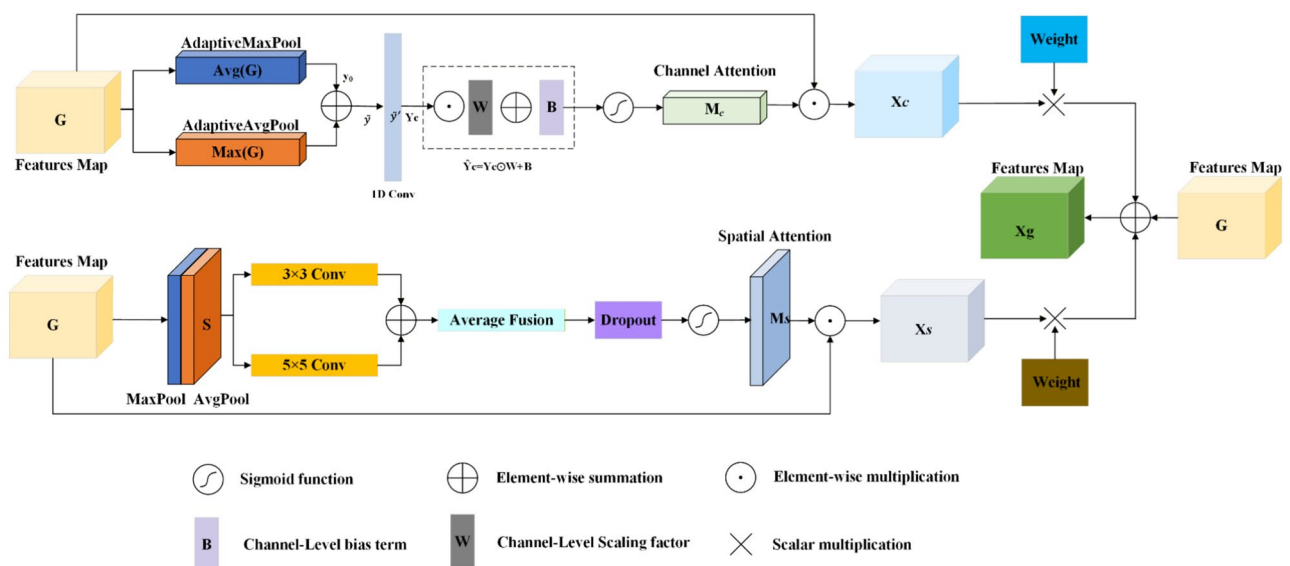


Fig. 8. Parallel multi-scale channel-spatial attention (PMS-CSA) module. Channel attention and spatial attention are computed in parallel, with the final three paths converging to produce attention-enhanced feature maps.

The $Avg(\mathbf{G})$ reflects the overall activation intensity of each channel, while $Max(\mathbf{G})$ emphasizes locally salient responses. In CXR images, these correspond to diffuse abnormalities and focal high-response regions, respectively. The two statistics are then summed to fuse complementary semantics:

$$\mathbf{y}_0 = Avg(\mathbf{G}) + Max(\mathbf{G}) \in \mathbb{R}^{B \times C_g \times 1 \times 1} \quad (10)$$

To model local channel correlations, the tensor is permuted:

$$\tilde{\mathbf{y}} = \text{permute}(\mathbf{y}_0) \in \mathbb{R}^{B \times 1 \times C_g} \quad (11)$$

A 1D convolution with kernel size 3 is applied:

$$\tilde{\mathbf{y}}' = \text{Conv1D}(\tilde{\mathbf{y}}) \in \mathbb{R}^{B \times 1 \times C_g} \quad (12)$$

where $\tilde{\mathbf{y}}'$ represents the feature enhanced with local channel dependencies. This operation is equivalent to imposing a local receptive field constraint along the channel dimension, enabling efficient modeling of inter-channel local dependencies with minimal parameters, thus avoiding redundancy introduced by high-dimensional fully connected layers. The tensor is then permuted back:

$$\mathbf{Y}_c = \text{permute}(\tilde{\mathbf{y}}') \in \mathbb{R}^{B \times C_g \times 1 \times 1} \quad (13)$$

where $\mathbf{Y}_c$ denotes the intermediate channel attention feature. Subsequently, learnable per-channel affine modulation parameters are introduced for linear recalibration:

$$\hat{\mathbf{Y}}_c = \mathbf{Y}_c \odot \mathbf{W} + \mathbf{B} \quad (14)$$

where $\odot$ denotes element-wise multiplication, and $\mathbf{W}$ and $\mathbf{B}$ represent channel-wise scaling and bias terms, initialized as 1 and 0, respectively. These parameters provide independent amplitude adjustment for each semantic channel, enhancing responses highly correlated with disease discrimination.

Finally, the channel attention map is obtained via a Sigmoid activation:

$$\mathbf{M}_c = \sigma(\hat{\mathbf{Y}}_c) \in (0,1)^{B \times C_g \times 1 \times 1} \quad (15)$$

where σ denotes Sigmoid activation. Applying it to the original feature:

$$\mathbf{X}_c = \mathbf{G} \odot \mathbf{M}_c \quad (16)$$

where $\mathbf{X}_c$ denotes the channel-enhanced feature map. This process adaptively strengthens discriminative semantic channels while maintaining a lightweight design, making it suitable for multi-label CXR classification tasks where different diseases rely on distinct high-level semantic channels. To further enhance the model's perception of lesion spatial distribution, the spatial attention module operates in parallel with channel attention. First, channel-wise statistical aggregation is performed:

$$S_{avg} = Avg_c(\mathbf{G}) \in \mathbb{R}^{B \times 1 \times H_g \times W_g}, S_{max} = Max_c(\mathbf{G}) \in \mathbb{R}^{B \times 1 \times H_g \times W_g} \quad (17)$$

The S_{avg} reflects global structural responses and S_{max} highlights potential abnormal regions, facilitating salient lesion localization under complex backgrounds. They are concatenated along the channel dimension:

$$\mathbf{S} = \text{concat}(S_{avg}, S_{max}) \in \mathbb{R}^{B \times 2 \times H_g \times W_g} \quad (18)$$

To simultaneously capture fine-grained texture variations and larger-scale structural information, parallel 3×3 and 5×5 convolutions are applied:

$$a_3 = \text{Conv}_{3 \times 3}(\mathbf{S}), a_5 = \text{Conv}_{5 \times 5}(\mathbf{S}) \quad (19)$$

Their outputs are equally fused:

$$a = \frac{a_3 + a_5}{2} \quad (20)$$

This multi-scale design captures both high-frequency local edge structures and large-area low-frequency shadow patterns in chest radiographs. To suppress spatial overfitting, Dropout2d is applied:

$$a' = \text{Dropout2d}(a) \quad (21)$$

The spatial attention map is then obtained via Sigmoid activation:

$$\mathbf{M}_s = \sigma(a') \in (0,1)^{B \times 1 \times H_g \times W_g} \quad (22)$$

Applying it to the original feature:

$$\mathbf{X}_s = \mathbf{G} \odot \mathbf{M}_s \quad (23)$$

where $\mathbf{X}_s$ denotes the spatially enhanced feature map. Learnable scalar parameters w_c and w_s are introduced for the channel and spatial attention paths, respectively, constrained by Sigmoid activation:

$$\omega_c = \sigma(w_c), \omega_s = \sigma(w_s) \quad (24)$$

They are normalized to ensure their sum equals 1:

$$\bar{\omega}_c = \frac{\omega_c}{\omega_c + \omega_s}, \bar{\omega}_s = \frac{\omega_s}{\omega_c + \omega_s} \quad (25)$$

The final fused output is:

$$\mathbf{X}_g = \bar{\omega}_c \cdot \mathbf{X}_c + \bar{\omega}_s \cdot \mathbf{X}_s + \mathbf{G} \quad (26)$$

where $\mathbf{X}_g$ denotes the global branch feature map enhanced by PMS-CSA. The first two terms correspond to channel-enhanced and spatial-enhanced features, while the third term is the identity mapping $\mathbf{G}$, preserving original structural information and maintaining gradient stability.

Similarly, $\mathbf{X}_l$ denotes the PMS-CSA-enhanced feature map of the local branch.

Cross-attention fusion module

This study utilizes cross-attention for dual-branch fusion, as illustrated in Fig. 9. The features from the two branches undergo spatial flattening, are transformed into sequence format, and are subsequently aligned to a unified dimension through linear mapping. The augmented global features act as queries, while the augmented local features serve as keys and values. Multi-head scaled dot-product attention facilitates cross-scale semantic interaction, thereby enabling the adaptive selection and reconstruction of critical local details under the guidance of global semantics. The resulting dual-branch collaborative representation is further refined through residual connections, layer normalization, and a feedforward network. Subsequently, it is reconstructed into a spatial feature representation to ensure spatial consistency with the global branch. To achieve a balance between local detail enhancement and global semantic stability, an adaptive gating fusion mechanism has been designed, as illustrated in Fig. 10. Initially, globally averaged pooling is applied to both the enhanced global branch features $\mathbf{X}_g$ and the enhanced local branch features $\mathbf{X}_l$, resulting in channel-level statistical representations $\mathbf{p}_g$ and $\mathbf{p}_l$. These representations are subsequently concatenated and input into a multilayer perceptron. A Sigmoid activation function produces an image-level scalar gating coefficient $\gamma \in (0,1)^{B \times 1}$. The gating coefficient is expanded to the spatial dimensions via a broadcasting mechanism and then used to perform a weighted combination of the fused features and the enhanced global features, enabling adaptive regulation of the dual-branch information. Importantly, no explicit positional encoding is introduced during the transformation of features from two-dimensional feature maps to sequence representations. Given that both branches are constructed using convolutional neural networks, their feature maps inherently preserve spatial structural

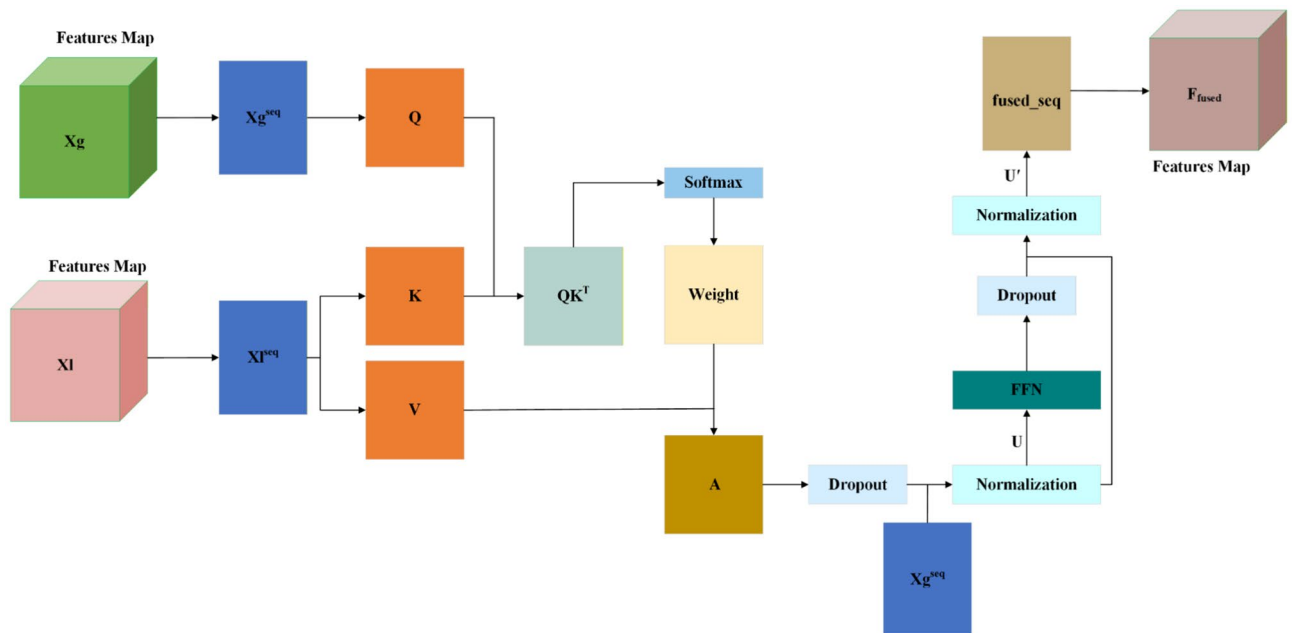


Fig. 9. Cross-Attention Fusion Module. This module generates feature maps that integrate global semantic information with local detail information.

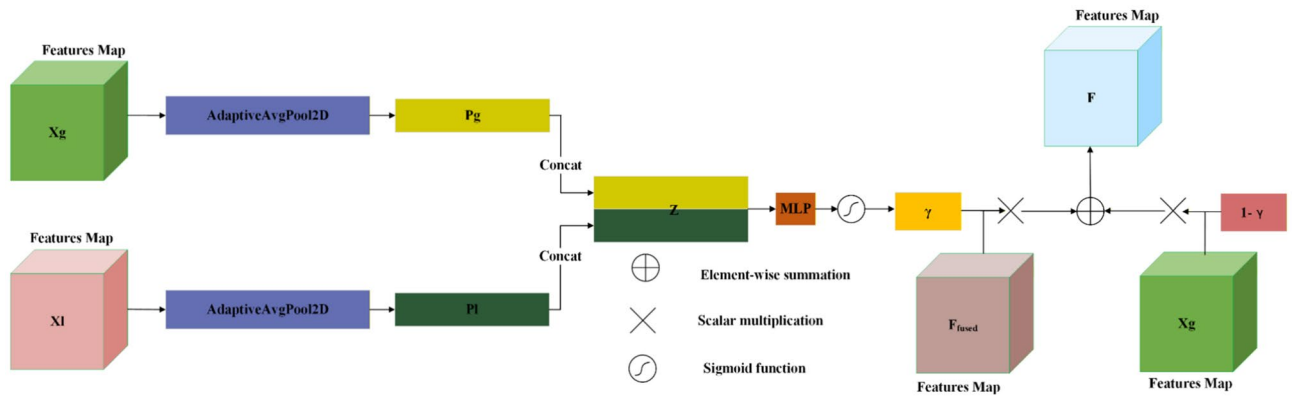


Fig. 10. Adaptive Gating Fusion Module. Dynamically selects the more important information source between the feature map resulting from combining global semantic and local detail information and the original global semantic feature map.

information. Moreover, the sequence flattening operation adheres strictly to the spatial arrangement order, thereby maintaining positional information. Following feature extraction, the network's high-dimensional outputs undergo adaptive global average pooling, which reduces the spatial dimensions to 1×1 and produces a global semantic representation. The features are then flattened and subjected to Dropout (0.4) to reduce the risk of overfitting. The classification head consists of a two-layer fully connected structure: the first connected layer transforms 1024 dimensional features into 512 dimensions, integrating Batch Normalization with ReLU activation functions to enhance both feature expressiveness and training stability. Subsequently, a Dropout rate of 0.5 is applied to further bolster model generalization. Finally, the second fully connected layer maps the 512-dimensional features to the number of classes (14 classes), generating the predicted results for each disease.

Formal mathematical description of the cross-attention fusion module

The enhanced global branch feature is denoted as $X_g \in \mathbb{R}^{B \times C_g \times H_g \times W_g}$, and the enhanced local branch feature as $X_l \in \mathbb{R}^{B \times C_l \times H_l \times W_l}$. First, the feature maps are flattened along the spatial dimensions and then permuted to form sequence representations:

$$X_g^{seq} = \text{permute}(\text{flatten}(X_g)) \in \mathbb{R}^{B \times N_g \times C_g}, X_l^{seq} = \text{permute}(\text{flatten}(X_l)) \in \mathbb{R}^{B \times N_l \times C_l} \quad (27)$$

where $N_g = H_g \times W_g$, $N_l = H_l \times W_l$. The operator $\text{flatten}(\cdot)$ denotes spatial flattening, and $\text{permute}(\cdot)$ denotes dimension permuting. The two sequences with a dimension of $d_{model} = 1024$ are used to construct the Query (Q), Key (K), and Value (V) matrices:

$$Q = X_g^{seq} W_q, K = X_l^{seq} W_k, V = X_l^{seq} W_v \quad (28)$$

where $W_q \in \mathbb{R}^{C_g \times d_{model}}$ and $W_k, W_v \in \mathbb{R}^{C_l \times d_{model}}$ are learnable parameter matrices., $Q \in \mathbb{R}^{B \times N_g \times d_{model}}, K, V \in \mathbb{R}^{B \times N_l \times d_{model}}$. The scaled dot-product attention is defined as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right)V \quad (29)$$

The model employs $h = 8$ attention heads, with each head having a dimension $d_h = 128$. For the i -th head:

$$\text{head}_i = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_h}}\right)V_i \quad (30)$$

Outputs from all heads are concatenated and linearly projected:

$$A = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_o \quad (31)$$

where $W_o \in \mathbb{R}^{d_{model} \times d_{model}}$. Dropout with rate 0.1 is applied within the attention layer to enhance training stability. This mechanism allows the query vectors generated by the global branch to guide the aggregation of local branch features, achieving cross-scale semantic modulation and enabling the model to adaptively select key local details under global structural constraints. After obtaining the dual-branch attention output, features are further refined via residual connections, layer normalization, and a feed-forward network (FFN):

$$U = \text{LayerNorm}(X_g^{seq} + \text{Dropout}(A)) \quad (32)$$

$$U' = \text{LayerNorm}(U + \text{Dropout}(\text{FFN}(U))) \quad (33)$$

LayerNorm denotes layer normalization, and $\mathbf{U}'$ denotes the sequence feature refined by both dual-branch attention and the FFN. The sequence feature is then reshaped back to a spatial feature map:

$$\mathbf{F}_{fused} = \text{unflatten}(\text{permute}(\mathbf{U}')) \in \mathbb{R}^{B \times d_{model} \times H_g \times W_g} \tag{34}$$

where $\mathbf{F}_{fused}$ denotes the dual-branch fused feature map integrating global semantics and local details. To balance local detail enhancement and global semantic stability, an adaptive gating fusion mechanism is designed. Global and local branch features are first pooled and flattened:

$$\mathbf{p}_g = \text{AdaptiveAvgPool2d}(\mathbf{X}_g) \xrightarrow{\text{flatten}} \mathbb{R}^{B \times C_g}, \quad \mathbf{p}_l = \text{AdaptiveAvgPool2d}(\mathbf{X}_l) \xrightarrow{\text{flatten}} \mathbb{R}^{B \times C_l} \tag{35}$$

where $\mathbf{p}_g$ and $\mathbf{p}_l$ denote channel-level semantic descriptor vectors for the global and local branches, respectively. The vectors are concatenated:

$$\mathbf{z} = [\mathbf{p}_g || \mathbf{p}_l] \in \mathbb{R}^{B \times (C_g + C_l)} \tag{36}$$

where $||$ denotes vector concatenation operations. And fed into a gated multi-layer perceptron (MLP) with a Sigmoid activation to obtain the gating coefficient:

$$\gamma = \sigma(\text{MLP}(\mathbf{z})) \tag{37}$$

where σ restricts the gating coefficient to the $(0,1)$ range, $\gamma \in (0,1)^{B \times 1}$. The final dual-branch fused output is computed as:

$$\mathbf{F} = \gamma \odot \mathbf{F}_{fused} + (1 - \gamma) \odot \mathbf{X}_g \tag{38}$$

with γ broadcasted to $\mathbb{R}^{B \times d_{model} \times H_g \times W_g}$, allowing the network to dynamically select the most informative contributions between the fused feature $\mathbf{F}_{fused}$ and the global semantic feature $\mathbf{X}_g$.

Loss function and two-stage training strategy

There are a large number of all-zero samples in the ChestX-ray14 dataset, and their proportion is significantly higher than that of any single disease category, resulting in a serious category imbalance during the training process. When directly optimizing with the complete training set, a large number of easily classifiable negative samples will dominate the loss calculation, causing the model to tend to output negative predictions for all disease labels, thereby weakening the discrimination ability for minority disease categories, especially low-incidence small lesions. To alleviate the above problems, this study proposes a two-stage progressive training strategy. In the first stage, we temporarily remove all zero samples, and the label space remains unchanged at 14 dimensions, enabling the model to prioritize learning discriminative features related to lesions. In this stage, the numerically stable BCEWithLogitsLoss is adopted as the optimization objective. In the second stage, a complete dataset is introduced, and a hybrid loss function is adopted. Based on binary cross-entropy, a three-layer mechanism of class-balanced weight, Focal modulation factor, and positive sample reweighting coefficient is applied successively at the category level, sample difficulty level, and label polarity level, thereby enhancing the model's learning ability for minority labels. To enhance experimental transparency, we further detail the number of images and patients at each stage, as illustrated in Table 1.

Formal mathematical description of the two-stage loss function

In the multi-label chest X-ray classification task, each image is associated with a binary label vector of length $C = 14$, $\mathbf{y} = [y_1, y_2, \dots, y_C]$, $y_i \in \{0,1\}$, where $y_i = 1$ indicates the presence of the i -th disease. After dual-branch feature extraction and multi-scale interaction fusion, the model outputs an unnormalized logit vector $\mathbf{z} = [z_1, z_2, \dots, z_C]$. The predicted probability for each class is obtained via the Sigmoid function:

$$\hat{y}_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}} \tag{39}$$

In the first stage, a numerically stable binary cross-entropy loss in logit form is adopted. The overall loss is computed as the mean over all classes:

$$\mathcal{L}_{BCE} = \frac{1}{C} \sum_{i=1}^C [\max(z_i, 0) - z_i y_i + \log(1 + e^{-|z_i|})] \tag{40}$$

Split	Stage1 images	Stage1 patients	Stage 2 images	Stage 2 patients
Train	36,754	10,080	78,964	21,563
Valid	4967	1441	11,509	3081
Test	10,038	2881	21,647	6161
All	51,759	14,402	112,120	30,805

Table 1. Statistics of images and patients across the two stages.

This formulation prevents gradient vanishing and numerical overflow in extreme probability regions, thereby ensuring stable convergence during lesion feature learning. In the second stage, a hybrid loss function is introduced to address class imbalance and the scarcity of minority samples. Built upon the standard binary cross-entropy loss, it integrates three mechanisms: class-balanced weight, Focal modulation factor, and positive-sample reweighting. For the i -th class, the standard binary cross-entropy loss is defined as:

$$\ell_{BCE,i} = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (41)$$

To mitigate class imbalance, a class-balanced weight is introduced:

$$w_i^{class} = \alpha y_i + (1 - \alpha)(1 - y_i) \quad (42)$$

The first-step weighted loss is:

$$\ell_i^I = w_i^{class} \cdot \ell_{BCE,i} \quad (43)$$

To reduce the dominance of easily classified samples, a Focal modulation factor is applied:

$$p_{t,i} = \begin{cases} \hat{y}_i, & y_i = 1 \\ 1 - \hat{y}_i, & y_i = 0 \end{cases} \quad (1 - p_{t,i})^\gamma, \gamma \geq 0 \quad (44)$$

where $p_{t,i}$ denotes the predicted probability corresponding to the ground-truth label of the i -th class. The second-step loss becomes:

$$\ell_i^{II} = (1 - p_{t,i})^\gamma \cdot \ell_i^I \quad (45)$$

To further enhance sensitivity to rare positive samples, a positive sample reweighting coefficient is introduced:

$$w_i^{pos} = 1 + (\omega - 1)y_i, \omega > 1 \quad (46)$$

The final hybrid loss for the i -th class is therefore:

$$\ell_i^{Hybrid} = w_i^{pos} \cdot \ell_i^{II} \quad (47)$$

which can be written in expanded form as:

$$\ell_i^{Hybrid} = [1 + (\omega - 1)y_i] (1 - p_{t,i})^\gamma [\alpha y_i + (1 - \alpha)(1 - y_i)] \ell_{BCE,i} \quad (48)$$

Averaging over all C classes yields the final second-stage hybrid loss:

$$\mathcal{L}^{Hybrid} = \frac{1}{C} \sum_{i=1}^C \ell_i^{Hybrid} \quad (49)$$

In the hybrid loss function, the hyperparameters are set as follows: class-balanced weight $\alpha = 0.5$, Focal modulation factor $\gamma = 2.0$, positive sample weighting coefficient $\omega = 1.5$. These parameters were determined via grid search on the validation set to achieve a balanced improvement in minority-class performance.

Results

Evaluation indicators

To comprehensively evaluate the performance of the proposed algorithm in multi-label classification tasks, we employed Area Under the Curve (AUC) and Receiver Operating Characteristic (ROC) curves as primary evaluation metrics. These metrics are particularly well-suited for addressing imbalanced label distributions in medical image analysis. Additionally, we calculated metrics including Sensitivity@Specificity, Specificity@Sensitivity, Sensitivity, Specificity, Accuracy and Area Under the Precision–Recall Curve (AUPRC) to further assess model performance.

Comparison results with baseline models

To ensure fairness, all comparison models were reimplemented using identical patient-level data splits of 7:1:2, unified preprocessing pipelines, and consistent training settings. To ensure a fair and controlled comparison, all baseline models were initialized with ImageNet-pretrained weights, consistent with the initialization strategy adopted for the two branches of the proposed model. The models included representative approaches such as classical CNNs like ResNet18 and DenseNet121³⁹, pure Transformer architectures such as Vision Transformer⁴⁰ and Swin Transformer⁴¹, as well as hybrid architectures like MaxViT⁴² and CoAtNet-2-rw. Table 2 presents the AUC and average AUC results for each method across 14 disease categories. The experimental findings indicate that the proposed CR-MSNet exhibits exceptional performance in multi-label chest radiograph classification, achieving an average AUC of 0.847, which exceeds that of the six benchmark models. Notably, CR-MSNet attains the highest AUC in 13 out of 14 disease categories. When compared to classical CNN networks, CR-MSNet enhances the average AUC by 4.7%, underscoring the significant contribution of the CoAtNet-2-rw global

Method	Atel	Cons	Infi	Pne1	Edem	Emph	Fibr	Effu	Pne2	PT	Card	Nodu	Mass	Hern	Mean
ResNet18	0.784	0.801	0.700	0.825	0.889	0.845	0.762	0.874	0.742	0.758	0.906	0.685	0.792	0.830	0.800
DenseNet121	0.816	0.811	0.711	0.864	0.902	0.915	0.815	0.889	0.762	0.779	0.908	0.780	0.850	0.841	0.831
Vision Transformer	0.797	0.807	0.708	0.846	0.894	0.881	0.794	0.874	0.742	0.754	0.893	0.734	0.815	0.796	0.810
Swin Transformer	0.809	0.817	0.715	0.867	0.895	0.903	0.807	0.885	0.760	0.781	0.910	0.770	0.848	0.821	0.828
MaxViT	0.799	0.811	0.710	0.851	0.895	0.893	0.795	0.877	0.756	0.769	0.895	0.761	0.819	0.814	0.818
CoAtNet-2-rw	0.824	0.816	0.709	0.868	0.901	0.918	0.817	0.891	0.767	0.780	0.919	0.794	0.859	0.829	0.835
CR-MSNet (ours)	0.827	0.822	0.721	0.882	0.907	0.930	0.826	0.893	0.761	0.801	0.921	0.802	0.876	0.883	0.847

Table 2. Quantitative comparison results on the ChestX-ray14 dataset. Atel, Cons, Infi, Pne1, Edem, Emph, Fibr, Effu, Pne2, PT, Card, Nodu, Mass, and Hern refer to the 14 disease classes, namely, Atelectasis, Consolidation, Infiltration, Pneumothorax, Edema, Emphysema, Fibrosis, Effusion, Pneumonia, Pleural_Thickening, Cardiomegaly, Nodule, Mass, and Hernia, respectively. Mean refers to the average AUC. Bold text indicates the best AUC for each disease class.

branch to overall classification performance. In comparison to a single Transformer backbone, the average AUC improves by 3.7%, further emphasizing the complementary role of the CNN local branch in capturing small lesions and fine-grained local features. Relative to traditional hybrid architectures, the improvement is 2.9%, illustrating the advantages of the Transformer backbone and ResNet18 parallel architecture in integrating global and local information. Compared to the main baseline algorithm, the average AUC increases by 1.2%, with a notable 5.4% enhancement for the minority disease Hernia. This demonstrates that the dual-branch architecture effectively amalgamates information, while the PMS-CSA attention module focus key lesions, significantly improving the ability to differentiate minority categories and small lesions.

From a structural perspective, a single CNN backbone excels at modeling local textures and edge information but is limited in capturing long-range dependencies. In contrast, a pure Transformer backbone offers advantages in global modeling but may struggle with fine-grained feature representation in small-scale or low-contrast lesion areas. Hybrid architectures attempt to integrate these strengths; however, they typically employ serial or shallow feature fusion strategies, leading to relatively limited cross-scale and dual-branch information interaction. In contrast, CR-MSNet utilizes a parallel dual-branch architecture to separately model global semantic and local detail information, while incorporating the PMS-CSA attention module to enhance responses in critical regions. Simultaneously, the cross-attention fusion mechanism facilitates deep interaction between dual-branch features, enabling the model to maintain global consistency while enhancing local structural representation capabilities. This collaborative modeling and deep fusion strategy improves model stability in multi-label imbalanced scenarios, yielding consistent performance gains across categories rather than relying on incidental improvements in individual categories. In summary, under uniform experimental conditions, CR-MSNet demonstrates consistent advantages in both average performance and category-level metrics, thereby validating the effectiveness of the dual-branch multi-scale attention framework for multi-label diagnosis tasks in chest imaging.

Classification results

The ROC curves are illustrated in Fig. 11. Overall, the model demonstrated high discriminative ability for most significant chest lesions, with Emphysema, Cardiomegaly, Edema, and Hernia exhibiting the best performance. This indicates that the model effectively recognizes lesions characterized by distinct structures and high contrast. Additionally, the model maintains robust stability for medium-scale lesions such as Pleural_Thickening, Pneumothorax, and Mass. In contrast, classes with pronounced nonspecific imaging features, including Infiltration and Pneumonia, continue to pose certain challenges. This study addresses a multi-label classification task, wherein the probability distributions and levels of imbalance differ across various disease categories. Consequently, category-independent decision thresholds are utilized rather than a single unified threshold. Specifically, following model training, the optimal decision threshold for each disease category is established based exclusively on the prediction outcomes from the validation set. This optimal threshold is derived by maximizing the F1 score on the validation set. The thresholds for the 14 disease categories are as follows: 0.42, 0.36, 0.44, 0.42, 0.38, 0.52, 0.38, 0.46, 0.85, 0.39, 0.45, 0.41, 0.41, and 0.33.

The remaining evaluation metric results are illustrated in Table 3. The model demonstrated high AUPRC values for diseases characterized by distinct structural features and consistent imaging presentations, including Effusion, Emphysema, and Mass. Notably, Effusion achieved an AUPRC of 0.550, indicating that the model retains good precision even at elevated recall intervals, effectively prioritizing positive samples in its predictions. Emphysema not only displays a high AUPRC but also maintains significant specificity under conditions of high sensitivity, with both Sensitivity and Specificity at elevated levels. This further corroborates the model's robust discrimination capability for lesions with well-defined boundaries. Cardiomegaly and Edema similarly exhibit relatively stable ranking capabilities and threshold performance. In contrast, categories characterized by semantically ambiguous boundaries or substantial labeling noise present ongoing challenges. For example, the AUPRC for pneumonia is merely 0.043, which approaches baseline levels under conditions of low prevalence. This finding indicates considerable overlap in the distributions of positive and negative sample predictions, thereby limiting the model's reliability in identifying positives within this category. On the other hand, Infiltration achieved an AUPRC of 0.377, reflecting an acceptable ranking capability but exhibiting low

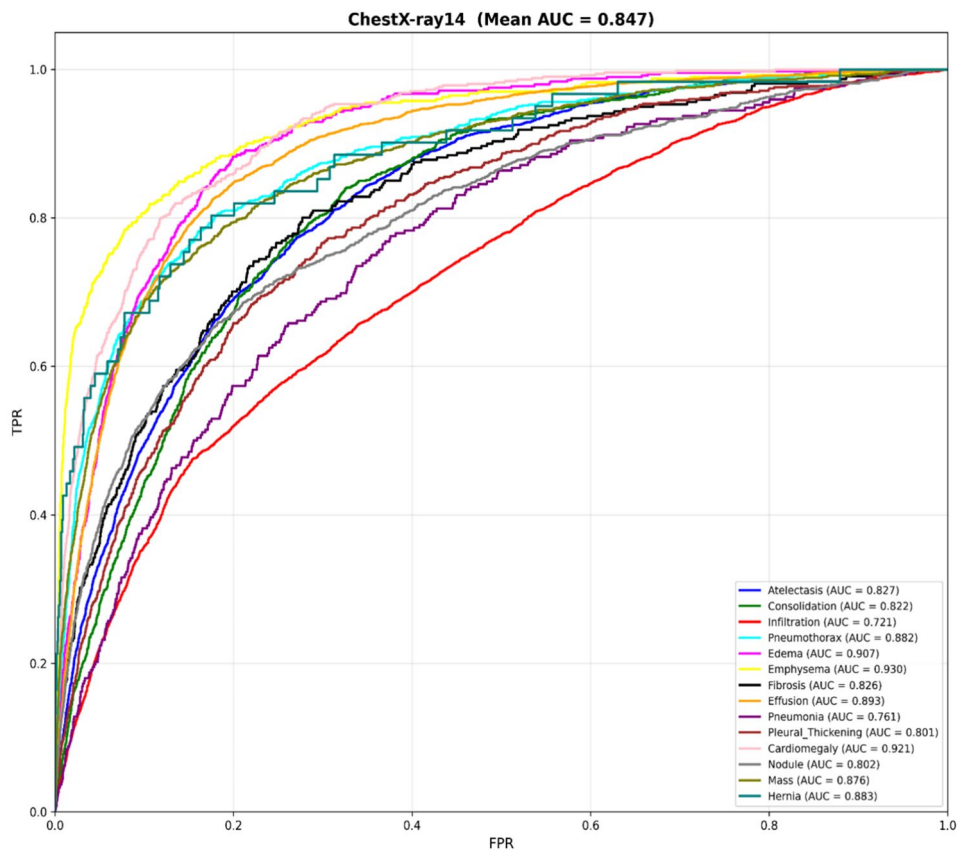


Fig. 11. ROC curves for CR-MSNet over 14 different diseases.

Disease	Sens@spec0.90	Spec@sens0.90	Sensitivity	Specificity	Accuracy	AUPRC
Atelectasis	0.495	0.558	0.773	0.730	0.734	0.367
Consolidation	0.442	0.573	0.840	0.675	0.682	0.160
Infiltration	0.358	0.309	0.568	0.756	0.722	0.377
Pneumothorax	0.689	0.631	0.806	0.817	0.817	0.340
Edema	0.703	0.768	0.883	0.801	0.803	0.214
Emphysema	0.806	0.785	0.819	0.894	0.893	0.388
Fibrosis	0.520	0.516	0.801	0.723	0.724	0.102
Effusion	0.689	0.720	0.828	0.821	0.822	0.550
Pneumonia	0.379	0.424	0.776	0.624	0.626	0.043
Pleural_Thickening	0.463	0.466	0.766	0.701	0.703	0.141
Cardiomegaly	0.758	0.761	0.820	0.865	0.864	0.315
Nodule	0.528	0.427	0.660	0.817	0.809	0.278
Mass	0.686	0.608	0.789	0.811	0.810	0.379
Hernia	0.672	0.634	0.803	0.825	0.825	0.194
Macro	0.585	0.584	0.781	0.776	0.774	0.275
Micro	0.660	0.697	0.837	0.789	0.792	0.369

Table 3. Additional evaluation metrics for CR-MSNet across 14 disease categories.

sensitivity under high specificity constraints (Sens@Spec0.90=0.358). This observation suggests that its recall capability significantly diminishes when clinical scenarios necessitate stringent control of false-positive rates. Consequently, some performance degradation may arise from intra-class phenotypic heterogeneity and the long-tail distribution of data, rather than from a complete absence of discriminatory ability in the model. The Macro AUPRC is 0.275, while the Micro AUPRC is 0.369. This indicates that the model exhibits more stable ranking capabilities in high-incidence categories with larger sample sizes, whereas low-incidence categories

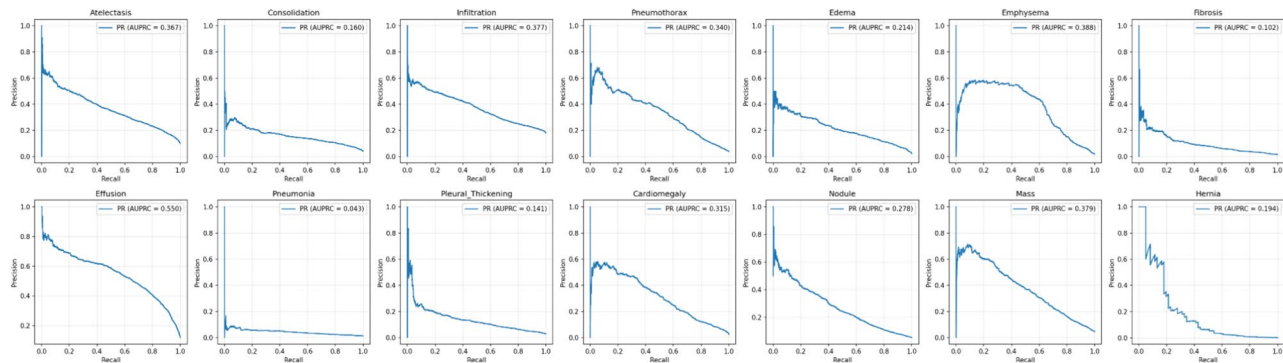


Fig. 12. Precision–Recall curves for the 14 disease categories on ChestX-ray14.

Disease	CR-MSNet AUC (95% CI)	CoAtNet-2-rw AUC (95% CI)	ΔAUC (95% CI)
Atelectasis	0.827(0.819,0.836)	0.824(0.815,0.832)	+0.003(+0.001,+0.006)
Consolidation	0.822(0.810,0.834)	0.816(0.804,0.829)	+0.006(+0.002,+0.010)
Infiltration	0.721(0.711,0.730)	0.709(0.699,0.718)	+0.012(+0.005,+0.019)
Pneumothorax	0.882(0.870,0.895)	0.868(0.856,0.881)	+0.014(+0.004,+0.024)
Edema	0.907(0.896,0.918)	0.901(0.890,0.912)	+0.006(+0.003,+0.009)
Emphysema	0.930(0.915,0.944)	0.918(0.903,0.932)	+0.012(+0.008,+0.017)
Fibrosis	0.826(0.803,0.849)	0.817(0.794,0.840)	+0.010(-0.002,+0.023)
Effusion	0.893(0.887,0.900)	0.891(0.885,0.898)	+0.002(-0.000,+0.004)
Pneumonia	0.761(0.733,0.789)	0.767(0.738,0.795)	-0.006(-0.021,+0.010)
Pleural_Thickening	0.801(0.784,0.818)	0.780(0.765,0.795)	+0.021(+0.009,+0.031)
Cardiomegaly	0.921(0.911,0.931)	0.919(0.907,0.928)	+0.001(-0.005,+0.008)
Nodule	0.802(0.788,0.815)	0.794(0.779,0.809)	+0.008(+0.003,+0.013)
Mass	0.876(0.864,0.888)	0.859(0.846,0.871)	+0.017(+0.007,+0.027)
Hernia	0.883(0.833,0.929)	0.829(0.779,0.880)	+0.054(+0.026,+0.085)
Mean	0.847(0.841,0.852)	0.835(0.830,0.840)	+0.012(+0.008,+0.016)

Table 4. Comparison of the 95% CI for the AUC metric between the CR-MSNet model and the CoAtNe-2-rw model, along with the 95% CI for ΔAUROC (CR-MSNet – CoAtNe-2-rw).

adversely affect the macro-average results. This discrepancy between macro and micro metrics reflects the long-tail distribution characteristic of medical imaging data. Additionally, the Macro Sensitivity, Macro Specificity, and Macro Accuracy are all approximately at the 0.78 level, suggesting that the model maintains relatively balanced classification performance at the overall decision-making level. Overall, the AUPRC values and the corresponding Precision–Recall (PR) curves (Fig. 12) demonstrate that the model achieves strong discriminative performance for high-prevalence diseases with distinct structural characteristics. However, its performance on low-prevalence and semantically complex categories remains relatively limited and necessitates further improvement.

Statistical analysis

To evaluate the statistical significance of performance enhancements, this study utilized patient-level nonparametric bootstrap methods⁴³ to estimate the 95% confidence interval (CI) for the AUC. In the test set, 1,000 resampling iterations with replacement were performed at the patient level, employing paired bootstrap techniques to preserve case correspondence between different models within the same resampled subset. For each resampled subset, AUC values for both CR-MSNet and CoAtNet-2-rw were computed concurrently, and their paired difference (ΔAUC = CR-MSNet – CoAtNet-2-rw) was documented. The 95% CI was derived from the distribution of differences obtained through resampling, utilizing the percentile method. The results are illustrated in Table 4, where the 95% CI for ΔAUC did not include zero in 10 out of 14 categories, indicating a statistically significant improvement in performance. Notable significant levels were observed in categories such as Pleural_Thickening (95% CI +0.009, +0.031), Mass (95% CI +0.007, +0.027), and Emphysema (95% CI +0.008, +0.017). In certain categories, including Pneumonia and Cardiomegaly, the CI for the difference encompassed zero, suggesting no significant performance difference between the two models. The overall mean AUC difference was +0.012 (95% CI +0.008, +0.016), with the interval excluding zero, which indicates that CR-MSNet demonstrates a statistically significant advantage over the primary baseline model, CoAtNet-2-rw, in the overall multi-label classification task. For most diseases, the CI widths for CR-MSNet clustered within the

range of 0.01 to 0.03, as seen in Effusion and Cardiomegaly, suggesting consistent discrimination capabilities for diseases with relatively abundant samples and stable imaging features. In contrast, diseases characterized by diffuse imaging manifestations and ambiguous semantic boundaries exhibited slightly wider 95% CIs for AUC, yet remained within acceptable limits overall, reflecting the model's robustness in the presence of complex lesion patterns. Hernia, a typical low-prevalence category, displayed the widest CI (0.833, 0.929). This observation aligns with the statistical characteristic that bootstrap resampling introduces greater performance variability when category samples are limited.

To visually represent the 95% CI for the AUC of the CR-MSNet model across 14 disease classes on the test set, we present the corresponding test results as bar charts with error bars in Fig. 13.

Qualitative analysis

This study utilizes Grad-CAM⁴⁴ to visualize the decision-making basis of CR-MSNet, thereby enhancing its interpretability. Grad-CAM produces category-specific activation heatmaps by leveraging gradient information from the feature maps of the final convolutional layer associated with target categories. This approach highlights the image regions that significantly influence the model's predictions. The spatial consistency analysis between the model-generated attention maps and the ground-truth bounding boxes was performed using the officially released bounding box subset of ChestX-ray14. As illustrated in Fig. 14, the red rectangular boxes on the original chest radiographs denote the lesion localization regions provided in the dataset, while the high-response areas in the corresponding heatmaps indicate the regions on which the model concentrates during classification. Colors that approach red signify a greater contribution of that region to the current category prediction. The displayed test subset samples reveal that, for certain disease categories—specifically Cardiomegaly, Nodule, Mass, Pneumothorax, and Infiltrate—the high-response regions generated by the model exhibit spatial overlap or proximity with the manually annotated lesion areas. This spatial consistency is primarily characterized by enhanced responses in the core regions of the lesions, rather than by precise pixel-level matching. It is crucial to emphasize that bounding box annotations are available for only a limited number of categories and samples. Consequently, the Grad-CAM results presented in this study serve as a qualitative interpretability analysis. They assist in demonstrating that the model concentrates on key disease-related regions during its decision-making process, rather than providing a systematic evaluation of localization performance.

To intuitively illustrate the class discrimination capability of CR-MSNet, this study selected twelve representative CXR prediction examples for analysis. The disease prediction scores are presented in Fig. 15. Overall, CR-MSNet effectively identified the primary lesions in most instances, particularly in categories with distinct imaging features such as Effusion, Mass, and Pneumothorax, where the prediction probabilities were generally significantly higher than those for other diseases. Furthermore, the ranking of predictions closely aligned with the true labels, underscoring the model's strength in detecting significant lesions across multiple scales. Additionally, for samples exhibiting multiple abnormalities, the network assigned high confidence to several true positives, demonstrating robust multi-label collaborative discrimination capability. However, the prediction results also highlighted certain challenges. For instance, in the cases depicted in the fourth row and fourth column, although the model accurately identified the true labels, its prediction scores were typically low for relatively blurred or low-contrast lesions such as Atelectasis, Fibrosis and Edema. Concurrently, the model

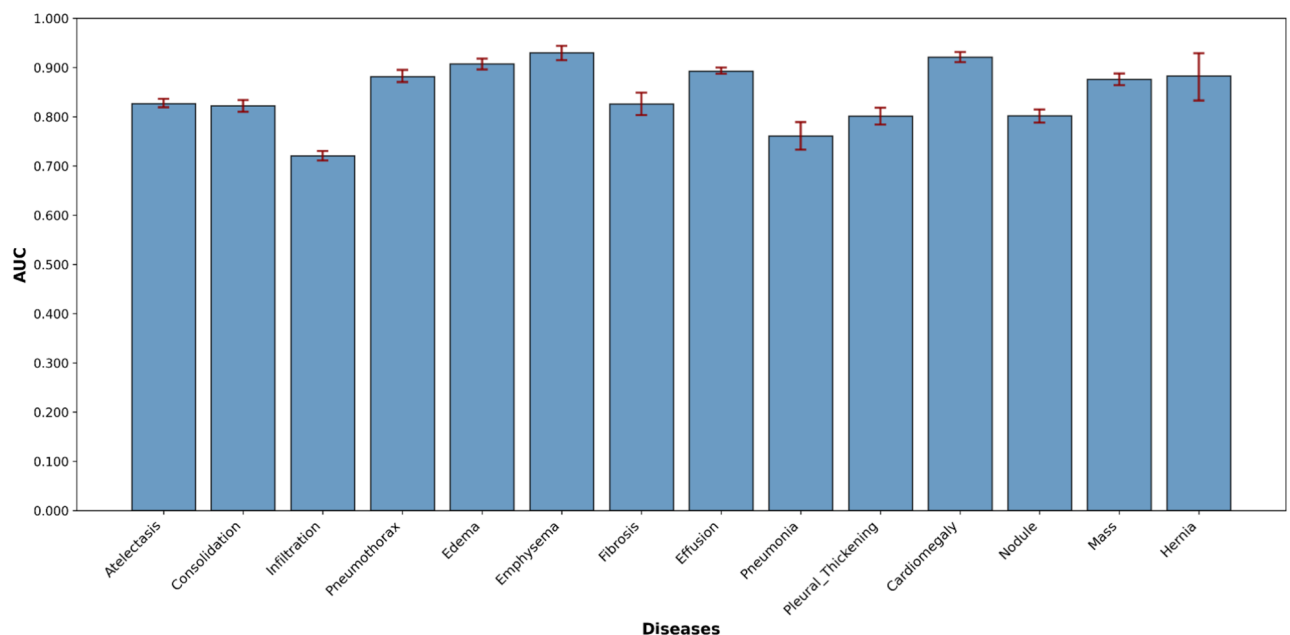


Fig. 13. The 95% CI for the AUC of the 14 disease classes in the test set was estimated using a nonparametric bootstrap method. The red error bars correspond to the 95% CI for the 14 disease classes.

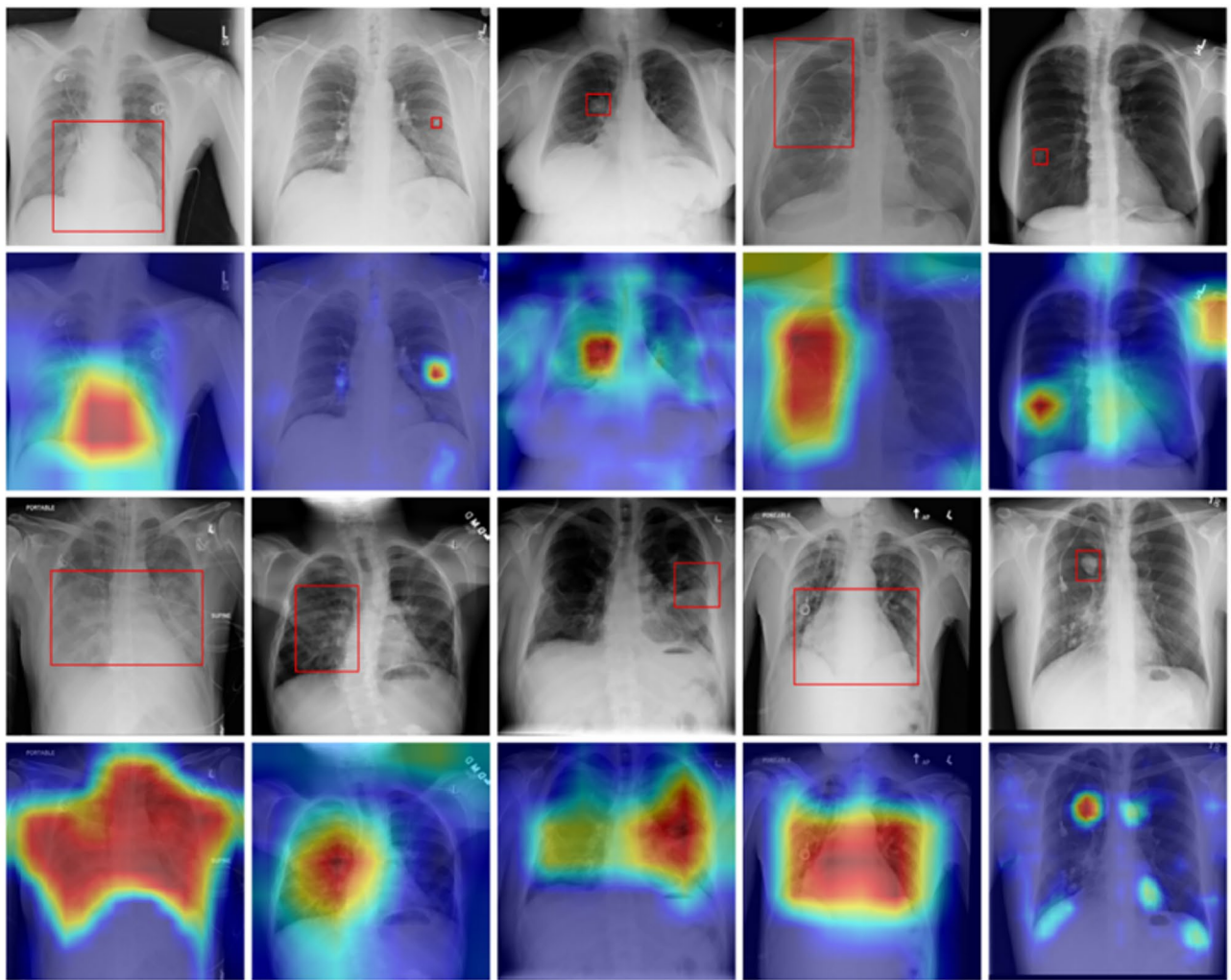


Fig. 14. Example of feature visualization based on Grad-CAM. The red rectangular boxes in the original chest radiographs denote lesion locations provided by the officially released bounding box subset of the ChestX-ray14 dataset. This subset contains 880 annotated images, which were divided into a training set (622 images), a validation set (116 images), and a test set (142 images). The spatial consistency analysis was conducted exclusively on the 142 annotated images in the test set.

assigned higher probabilities to Pleural Thickening, Nodule, and Consolidation, which were not true positives, indicating some confusion and instability when addressing fine-grained lesions and similar classes. In summary, the example prediction results indicate that CR-MSNet can reliably extract essential lesion information and provide dependable class probability distributions in complex chest radiographs.

Cross-dataset generalization analysis

To assess the model's cross-dataset generalization capability, we performed external validation using the CheXpert dataset⁴⁵, which consists of 224,316 chest X-ray images from 65,240 patients. In contrast to ChestX-ray14, CheXpert presents notable differences in image resolution, patient demographics, and labeling strategies, thus creating a distribution-shift scenario. This study did not involve any fine-tuning or retraining on CheXpert. The CR-MSNet was directly evaluated on the CheXpert validation subset (without uncertainty labels), which consists of 234 CXRs. To ensure a fair comparison, only frontal chest radiographs were included. Evaluation focused exclusively on overlapping categories with consistent label definitions across both datasets: Atelectasis, Cardiomegaly, Pneumonia, Pneumothorax, Consolidation, and Edema. The results are in Table 5, CR-MSNet achieved an average AUC of 0.808 on CheXpert, a decline from the 0.853 AUC recorded on ChestX-ray14. This variation in performance may be attributed to differences in data distribution, labeling methodologies, and patient populations. Despite these distribution shifts, the model exhibited relatively stable diagnostic performance, suggesting that the proposed method retains a degree of generalization capability across datasets.

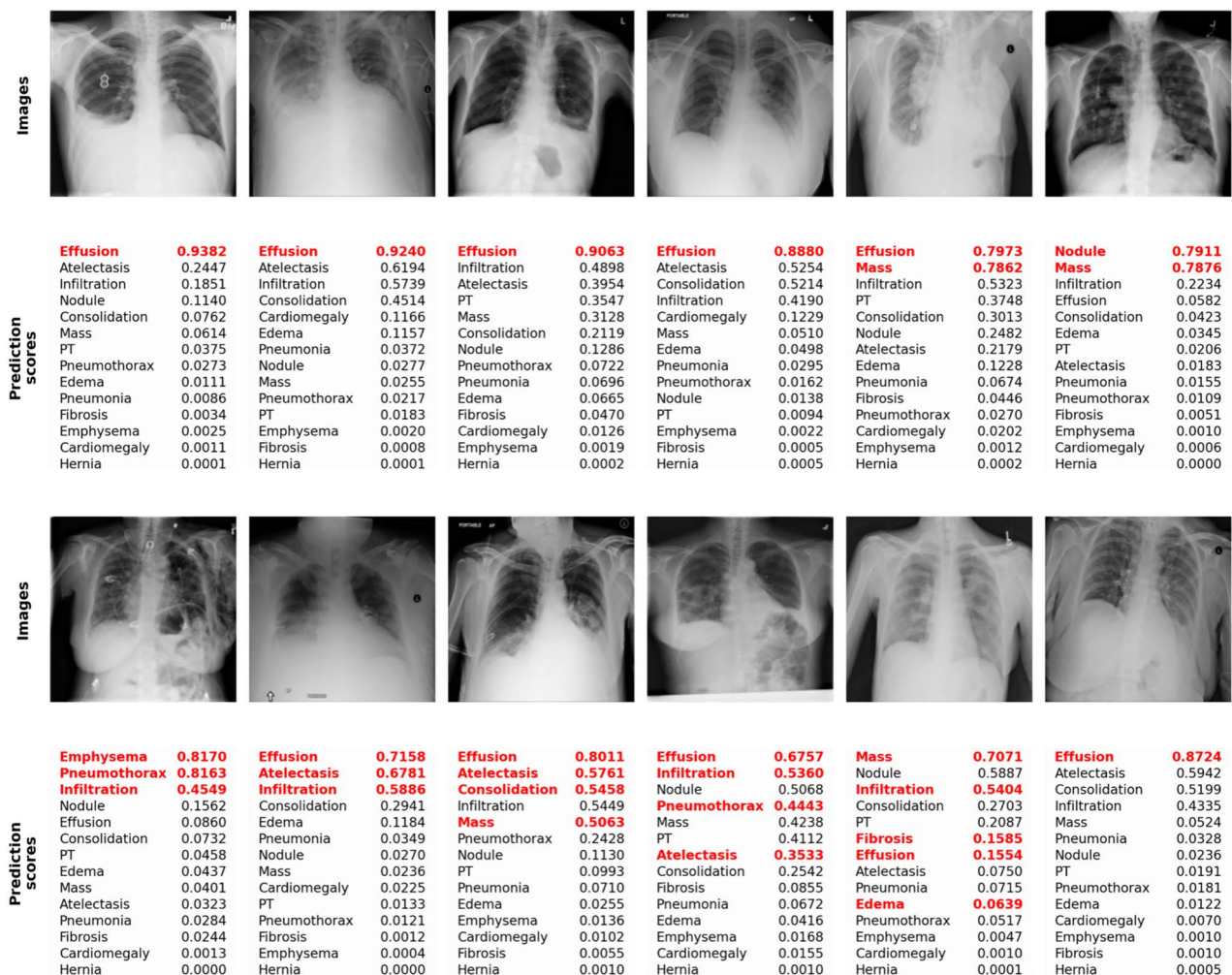


Fig. 15. Visualization example of prediction results for test set samples. The figure displays the model's prediction scores for selected disease classes, sorted by confidence level from highest to lowest; the true labels for each sample are annotated in red font. This visualization intuitively reflects the prediction distribution characteristics of the model in the multi-label chest X-ray image classification task.

Disease	ChestX-ray14	CheXpert
Atelectasis	0.827	0.831
Cardiomegaly	0.921	0.857
Pneumonia	0.761	0.724
Pneumothorax	0.882	0.760
Consolidation	0.822	0.876
Edema	0.907	0.800
Mean	0.853	0.808

Table 5. Comparison of AUC and Mean AUC for CR-MSNet on overlapping disease categories in the ChestX-ray14 and CheXpert datasets.

Efficiency analysis

This study systematically evaluates the efficiency of the proposed CR-MSNet from three perspectives: (1) model parameters, (2) computational complexity (FLOPs), and (3) inference latency. The parameter results are illustrated in Table 6, indicating that CR-MSNet comprises 114.60 M parameters and necessitates 17.56G FLOPs, which exceed those of the single-backbone ResNet18 and the hybrid CoAtNet-2-rw model. This increase is primarily due to the dual-branch multi-scale feature modeling architecture. In terms of performance, CR-MSNet achieves a mean AUC of 0.847, surpassing ResNet18 and CoAtNet-2-rw by 4.7% and 1.2%, respectively.

Model	Parameters	Flops	Inference time (ms)	Mean
Resnet18	24.56 M	4.13G	7 ms	0.800
CoAtNet-2-rw	73.38 M	14.48G	18.6 ms	0.835
CR-MSNet	114.60 M	17.56G	25.8 ms	0.847

Table 6. Comparison of efficiency analysis among CR-MSNet and its two constituent branches, ResNet18 and CoAtNet-2-rw.

These findings demonstrate that multi-scale collaborative modeling consistently yields stable performance improvements. Although the model size has increased, the enhancement in predictive performance corresponds to the additional computational cost, indicating that the architectural design provides effective and meaningful returns. Regarding inference efficiency, CR-MSNet requires 25.8 ms per image in a GPU environment, resulting in a theoretical throughput of approximately 38 images per second. In typical clinical scenarios, this latency is significantly lower than the time needed for manual interpretation, suggesting that the proposed method can meet real-time or near-real-time computer-aided diagnosis requirements when supported by standard GPU hardware. Overall, the proposed approach achieves a reasonable trade-off between performance enhancement and computational efficiency.

Calibration analysis

In addition to assessing discriminative capability, we evaluate the model’s performance in probabilistic calibration. To measure the alignment between predicted probabilities and actual label frequencies, we utilize the Expected Calibration Error (ECE)⁴⁶ as our evaluation metric, reporting the macro-average ECE across all categories. Specifically, within each category, we partition the predicted probabilities into equal-width bins. We then compute the difference between predicted confidence and true accuracy for each bin, averaging these differences across all categories. In the absence of calibration, the final model demonstrates a notable degree of overconfidence (Macro ECE=0.498), indicating that predicted probabilities are generally inflated. This overconfidence is linked to class reweighting and the Focal modulation mechanism, which enhances gradient responses for positive samples—an issue commonly observed in long-tail multi-label learning tasks. The underlying concern is one of probability scale bias rather than a reduction in category ranking capability. To enhance the reliability of probability outputs in clinical decision-making, we implemented category-level linear post-processing adjustments to the predicted probabilities for each category within the validation set, guided by principles of posterior calibration⁴⁶. Specifically, we executed independent linear scaling and bias correction on the outputs of each category in the probability space to reduce overconfidence and improve the alignment between predicted probabilities and true positive rates. Calibration parameters were derived exclusively from the validation set and were fixed during evaluation on the test set. Following calibration, the Macro ECE significantly decreased to 0.004 from its uncalibrated state, while the AUC remained unchanged. This finding indicates that the post-processing did not affect the model’s discriminative ranking capability but rather recalibrated the confidence levels. This result illustrates that, despite a tendency toward overconfidence in the uncalibrated state, the probability outputs from the two-stage training strategy demonstrate good calibrability. Thus, category-level linear post-processing can effectively enhance probability credibility while preserving high discriminative performance.

Ablation experiments
Ablation study of each module

To ensure the fairness of the ablation experiments, all models in the second stage employed the pre-trained weights obtained from the first stage. In the first stage, more lesion-sensitive feature representations were achieved by excluding all-zero samples. Consequently, the ablation experiments in the second stage were conducted using the same stable initialization to minimize interference from variations in training strategies. In the second stage, we preserved the complete label space and maintained consistent hybrid loss function settings, systematically evaluating the independent contributions of each component to the final performance by progressively incorporating them. The ablation results are illustrated in Table 7, with a detailed analysis as follows:

- (1) CoAtNet-2-rw: Utilizing CoAtNet-2-rw as the backbone, the average AUC achieves 0.835, reflecting robust overall performance in CXR classification, particularly for large-scale and pronounced structural abnormalities such as cardiomegaly and edema. Nevertheless, CoAtNet-2-rw exhibits limitations in detecting small-scale lesions, including hernia and mass, as well as in identifying blurred boundaries or weak textures, such as infiltration, and fine-grained texture lesions like fibrosis. This limitation primarily arises from its global attention mechanism and downsampling process, which may diminish the detection of small local signals and fine-grained textures.
- (2) ResNet18: When ResNet18 is utilized as the sole backbone, the average AUC declines to 0.800, which is generally lower than that of CoAtNet-2-rw across 14 disease classes. Notably, classes such as Nodule, Emphysema, and Mass show particularly significant reductions. This finding indicates that pure convolutional networks, which lack global semantic enhancement, struggle to capture signals that require global context in chest radiographs. Therefore, while ResNet18 exhibits strengths in certain local textures, it is insufficient as a singular backbone for this task.

Module	Atel	Cons	Infi	Pne1	Edem	Emph	Fibr	Effu	Pne2	PT	Card	Nodu	Mass	Hern	Mean
CoAtNet-2-rw	0.824	0.816	0.709	0.868	0.901	0.918	0.817	0.891	0.767	0.780	0.919	0.794	0.859	0.829	0.835
ResNet18	0.784	0.801	0.700	0.825	0.889	0.845	0.762	0.874	0.742	0.758	0.906	0.685	0.792	0.830	0.800
CoAtNet-2-rw + ResNet18	0.825	0.816	0.716	0.879	0.897	0.920	0.818	0.889	0.777	0.780	0.917	0.792	0.860	0.830	0.837
CoAtNet-2-rw + ResNet18 + PMS-CSA	0.832	0.822	0.724	0.877	0.908	0.928	0.832	0.890	0.776	0.787	0.921	0.797	0.872	0.840	0.843
CR-MSNet	0.827	0.822	0.721	0.882	0.907	0.930	0.826	0.893	0.761	0.801	0.921	0.802	0.876	0.883	0.847

Table 7. Comparison of ablation results. Significant values are in bold. Bold text indicates the best AUC for each disease class.

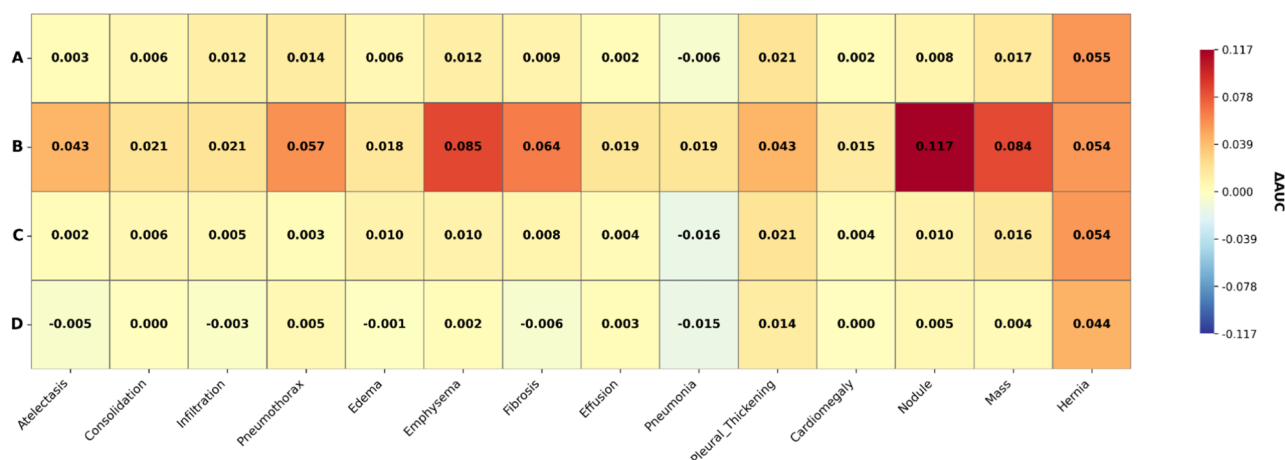


Fig. 16. The final model CR-MSNet achieved AUC/maAUC gains over four ablation variants across 14 types of chest diseases. A is the CoAtNet-2-rw model, B is the ResNet18 model, C is the CoAtNet-2-rw + ResNet18 model with concatenation and 1×1 convolution for fusion, and D is the CoAtNet-2-rw + ResNet18 + PMS-CSA model.

- CoAtNet-2-rw + ResNet18: By establishing a parallel structure with CoAtNet-2-rw and ResNet18, and employing concatenation followed by a 1×1 convolution for fusion, the average AUC marginally increases to 0.837. Most disease metrics exhibit slight improvements or remain stable, suggesting that the parallel dual-branch architecture surpasses the performance of the single-branch structure. This architecture can integrate global semantics and local textures to a certain degree; however, the straightforward concatenation-based fusion yields only modest gains, falling short of fully exploiting the complementary potential of the two branches.
- CoAtNet-2-rw + ResNet18 + PMS-CSA: Following the integration of the PMS-CSA module into each branch, the average AUC increased to 0.843. Both Infiltration and Fibrosis exhibited improvements of 0.015 relative to the baseline CoAtNet-2-rw, while Mass showed an enhancement of 0.013. These results indicate that the spatial attention and channel selectivity of PMS-CSA significantly improved the detection of small-scale, weakly textured, and fine-grained lesions, addressing the limitations of CoAtNet-2-rw in these categories. In conclusion, PMS-CSA effectively mitigates the backbone's deficiencies in local feature extraction, demonstrating that our proposed attention module can adeptly adapt to the intricate structures present in chest radiographs.
- CR-MSNet: After employing the cross-attention fusion module in the parallel two-branch fusion stage, the final CR-MSNet model achieved an average AUC of 0.847, the highest among all variants, representing an improvement of 0.012 over the baseline CoAtNet-2-rw. This result indicates that our dynamic balancing of fused features with the original global features is effective, surpassing the performance of a simple concatenation followed by a convolution with a 1×1 kernel. This approach enables the model to maintain both global semantic consistency and local structural sensitivity, particularly for Hernia, Mass, Pleural-Thickening and Pneumothorax.

We conducted a further analysis of the AUC gain distribution for the final model, CR-MSNet, in comparison to the four ablation variants across 14 disease classes. The AUC gain distribution is illustrated in Fig. 16. Each module's contribution to different diseases is visualized, with red indicating an increase in AUC and blue indicating a decrease. The figure demonstrates that the final model achieved varying degrees of performance improvement compared to the models at each stage across all disease classes. Notably, the incorporation of

Module	Atel	Cons	Infi	Pne1	Edem	Emph	Fibr	Effu	Pne2	PT	Card	Nodu	Mass	Hern	Mean
SE	0.819	0.812	0.708	0.866	0.899	0.914	0.806	0.889	0.765	0.777	0.919	0.790	0.859	0.829	0.832
CBAM	0.823	0.817	0.716	0.875	0.898	0.921	0.821	0.892	0.767	0.784	0.918	0.801	0.864	0.800	0.836
ECA	0.824	0.816	0.715	0.874	0.900	0.922	0.817	0.892	0.770	0.784	0.921	0.799	0.863	0.810	0.836
PMS-CSA	0.827	0.822	0.721	0.882	0.907	0.930	0.826	0.893	0.761	0.801	0.921	0.802	0.876	0.883	0.847

Table 8. Comparison of PMS-CSA with SE, CBAM, and ECA under identical network settings. Significant values are in bold. Bold text indicates the best AUC for each disease class.

Module	Atel	Cons	Infi	Pne1	Edem	Emph	Fibr	Effu	Pne2	PT	Card	Nodu	Mass	Hern	Mean
CA	0.826	0.820	0.719	0.880	0.900	0.920	0.825	0.889	0.765	0.797	0.917	0.790	0.865	0.875	0.842
SA	0.825	0.821	0.717	0.880	0.906	0.924	0.832	0.893	0.772	0.789	0.913	0.801	0.868	0.857	0.843
SA + CA	0.830	0.823	0.716	0.883	0.901	0.924	0.827	0.891	0.777	0.791	0.920	0.801	0.871	0.844	0.843
PMS-CSA	0.827	0.822	0.721	0.882	0.907	0.930	0.826	0.893	0.761	0.801	0.921	0.802	0.876	0.883	0.847

Table 9. Comparison of AUC per class and mean AUC across different models in modular contributory ablation studies. Significant values are in bold.

PMS-CSA significantly enhanced the model’s capacity to characterize weak textures and multi-scale features. This visualization offers an intuitive understanding of the specific contributions of each module to the model’s performance enhancement, thereby validating the effectiveness of the proposed structural design.

Comparison of different attention mechanisms

To assess the novelty of PMS-CSA, we substituted it with classical attention modules (SE, CBAM, and ECA) while maintaining the same network architecture and training conditions for a fair comparison. The results are detailed in Table 8. PMS-CSA surpasses existing attention modules in terms of mean AUC and across most disease categories, achieving an average enhancement of approximately 1.1–1.5%. These improvements are particularly pronounced for small-lesion or minority classes, further illustrating its capacity to effectively highlight discriminative channels and critical spatial regions, thereby affirming the significant innovation of the proposed module.

Ablation study of the PMS-CSA module

To systematically evaluate the effectiveness of the proposed PMS-CSA module in channel attention module, spatial attention module, and fusion module, this study established three control settings: one retaining only the channel attention module (CA), another retaining only the spatial attention module (SA), and a third that fuses CA and SA through element-wise multiplication (SA + CA). The experimental results are illustrated in Table 9.

- (1) CA module: The average AUC attains a value of 0.842. This module enhances semantically sensitive channels by employing channel statistical modeling and capturing local dependencies, thereby endowing the network with robust channel selection capabilities. Overall, the CA demonstrates consistent improvements across the majority of classes; however, the absence of spatial saliency modeling restricts its ability to fully represent features associated with dependency morphology, regional distribution, or local structure, consequently limiting the extent of enhancement.
- (2) SA module: The average AUC increases to 0.843. The SA module simultaneously captures local and large-scale spatial structures through multi-scale convolutional kernels, thereby enhancing the network’s ability to identify spatial patterns and regional variations. This approach compensates for the structural sensitivity that CA fails to deliver in certain classes. Nevertheless, SA remains limited in its capacity to select high-dimensional features, which hinders its performance in classes characterized by high channel redundancy or significant cross-channel semantic correlation.
- (3) CA + SA module: The average AUC is maintained at 0.843. While this method incorporates both channel and spatial information, element-wise multiplication represents a highly constrained approach that necessitates a significant degree of consistency in the attention responses of the two methods. If this consistency is lacking, it results in excessive feature suppression. The findings indicate that this hard-coupling method fails to consistently leverage the complementarity of the two types of attention in certain classes, with the improvement being largely comparable to that achieved by a single module.

In contrast, the complete PMS-CSA facilitates flexible collaboration among modules through a learnable three-path adaptive fusion strategy, achieving the highest average AUC of 0.847. This architecture dynamically balances channel enhancement, spatial saliency enhancement, and preservation of the original structure, thereby enabling the model to generate more robust representations across various feature levels and effectively mitigating information decay associated with simple multiplication. Additionally, the dual-branch insertion strategy further improves the multi-scale feature fusion capability, resulting in more balanced overall performance across classes.

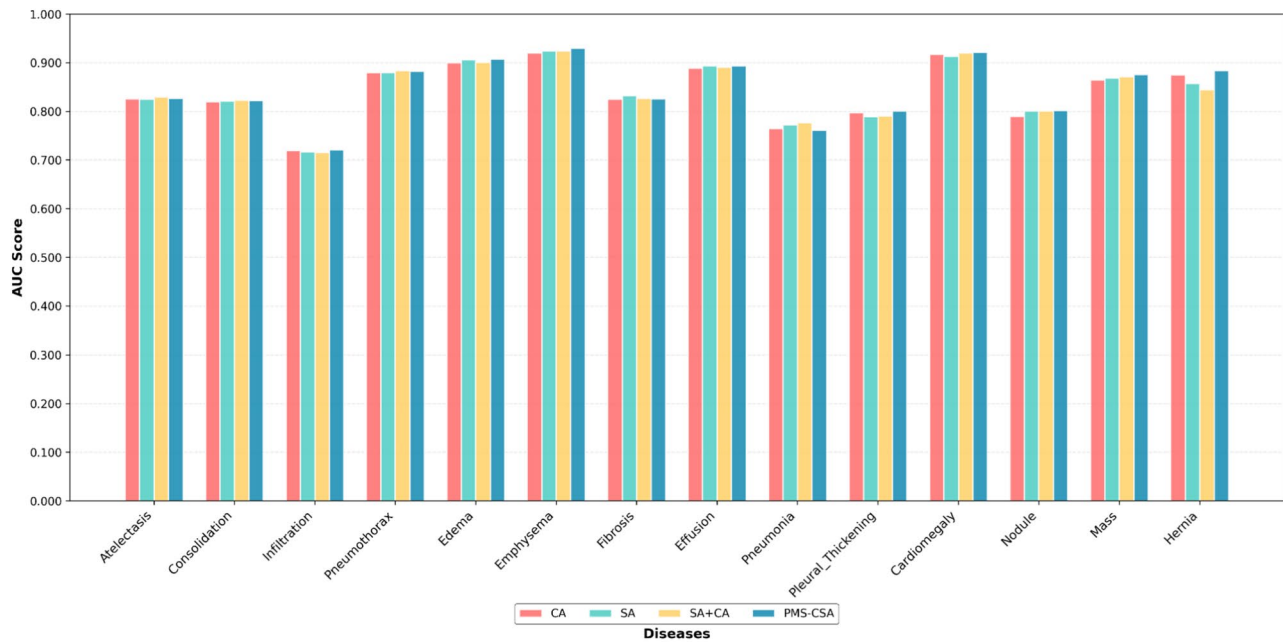


Fig. 17. Per-class AUC comparison of different models in the ablation study on module contributions.

Method	Atel	Cons	Infi	Pne1	Edem	Emph	Fibr	Effu	Pne2	PT	Card	Nodu	Mass	Hern	Mean
Full BCEWL	0.796	0.814	0.710	0.839	0.892	0.850	0.769	0.881	0.755	0.762	0.906	0.710	0.826	0.806	0.808
Full Hybrid	0.794	0.813	0.708	0.840	0.892	0.851	0.779	0.880	0.740	0.771	0.907	0.708	0.816	0.832	0.809
Filter BCEWL	0.804	0.808	0.616	0.870	0.896	0.917	0.806	0.887	0.720	0.766	0.915	0.777	0.858	0.769	0.815
Filter Hybrid	0.807	0.805	0.620	0.865	0.893	0.917	0.812	0.885	0.736	0.773	0.911	0.770	0.843	0.826	0.819
Two stage	0.827	0.822	0.721	0.882	0.907	0.930	0.826	0.893	0.761	0.801	0.921	0.802	0.876	0.883	0.847

Table 10. Comparison of AUC and Mean AUC for 14 disease categories under different training strategies. Bold text indicates the best AUC for each disease class.

To clearly illustrate the ablation effects of each component of the attention module, we present the verification results of four control settings in Fig. 17.

Two-stage training strategy

To systematically evaluate the independent and synergistic effects of filtering mechanisms and loss reweighting on overall performance and the recognition capability of minority class labels, we conducted ablation experiments that compared adjustments across two dimensions: data distribution and loss function. We examined the following methods: Full BCEWL (Full Data + BCEWLwithLogitsLoss), Full Hybrid (Full Data + Hybrid Loss), Filter BCEWL (Training set filtered for all-zero samples + BCEWLwithLogitsLoss), Filter Hybrid (Training set filtered for all-zero samples + Hybrid Loss), Two-stage (ours). The first four methods utilize a single-stage training strategy, while this study implements a progressive two-stage training approach. The results are illustrated in Table 10.

- (1) Sample filtering alone increases the average AUC from 0.808 to 0.815, showcasing that without loss reweighting, reducing interference from all-zero samples effectively alleviates the negative sample dominance issue. Particularly notable is the enhancement in performance for minority diseases such as Fibrosis and Emphysema, indicating that sample filtering boosts the model's capability to differentiate minority samples.
- (2) Under the complete data distribution, the introduction of a mixed loss mechanism did not yield a significant improvement in the average AUC. Nevertheless, it did enhance classification performance for certain minority diseases, such as Hernia and Fibrosis. This finding suggests that the mixed loss mechanism can partially alleviate class imbalance and improve the model's capacity to differentiate minority samples. However, when operating under the original distribution, which is predominantly characterized by extreme negative samples, reliance solely on loss reweighting does not substantially enhance overall performance. This indicates that bias at the data distribution level continues to limit the model's optimization potential.
- (3) The introduction of a mixed loss function within the context of the filtered data distribution resulted in a performance improvement to 0.819, significantly enhancing the AUC for diseases affecting minority groups. For example, the AUC for Hernia increased to 0.826, while that for Mass rose to 0.843. These findings demonstrate the synergistic effect of the filtering mechanism and loss reweighting: filtering reduces

the influence of negative samples, whereas the hybrid loss function improves the model's discriminative capability for minority samples.

- (4) In contrast, the proposed two-stage strategy attains the highest performance, evidenced by an average AUC of 0.847, along with significant enhancements in the AUCs for minority diseases, including Emphysema, Mass, and Hernia. This finding illustrates that the dynamic distribution adjustment and staged optimization mechanism can work synergistically to optimize both “early feature enhancement” and “late-stage distribution alignment.” As a result, this approach not only elevates overall performance but also markedly enhances recognition capabilities for long-tail categories.

Discussion

This study proposes a dual-branch multi-scale attention network, CR-MSNet, which consistently surpasses baseline models across numerous disease classes, showcasing notable performance advantages and robustness. The identification of chest abnormalities hinges on a dual comprehension of local details and global semantics. Traditional single-branch networks often fail to capture both local lesion details and global semantic information simultaneously. CR-MSNet utilizes CoAtNet-2-rw as the global branch for semantic modeling. Its progressive design of “pre-convolution, post-attention” effectively merges the benefits of convolutional operators and self-attention mechanisms at different depth stages, facilitating the capture of long-range dependent features across regions. The local branch, ResNet18, concentrates on local feature extraction, adeptly modeling local anatomical structures and fine-grained lesion patterns in chest X-rays through layer-by-layer residual blocks and feature enhancement modules. This dual-branch architecture integrates features at varying scales, achieving a unified modeling of lesions across multiple scales and significantly enhancing the capacity to identify both small local lesions and globally distributed lesions.

The PMS-CSA module implemented in each branch further enhances feature representation. Chest radiographs exhibit complex structures and varying lesion scales, which complicates the ability of traditional convolutional features to concentrate on critical regions. The channel attention module produces complementary channel descriptions through global average pooling and global max pooling, recalibrating them with learnable affine modulation parameters to strengthen discriminative semantic channels. The spatial attention module captures multi-scale spatial information via 3×3 and 5×5 parallel convolutions, while a three-path fusion strategy harmonizes channel enhancement, spatial enhancement, and original identity mapping information to facilitate efficient information integration. This module significantly refines the dual-branch output feature map, enhancing the focus on critical lesion regions and notably improving the recognition performance of small lesions. Subsequently, a dual-branch cross-attention mechanism enables adaptive interaction between global and local features, allowing the model to fully exploit complementary information across branches, optimize overall performance, and preserve spatial layout and semantic consistency. CR-MSNet employs a two-stage training strategy that effectively addresses the class imbalance issue present in the ChestX-ray14 dataset by utilizing different loss functions. The second stage incorporates the complete dataset to more accurately simulate clinical application scenarios.

Overall, CR-MSNet effectively balances multi-scale feature modeling with the management of class imbalance. The dual-branch architecture facilitates a more thorough acquisition of lesion information, while the attention module improves adaptability to the intricate structures present in chest radiographs. Additionally, the two-stage training strategy reinforces the model's robustness in real-world clinical settings. Further calculations of the 95% CI on the test set confirm the reliability and stability of the model's predictions, offering a viable solution for the multi-label classification of clinical chest radiographs.

Limitations and future work

Although CR-MSNet demonstrated competitive performance on ChestX-ray14, several limitations persist. The model excels in detecting globally distributed or structural abnormalities, such as emphysema, cardiomegaly, and edema, but is less effective for diseases characterized by blurred boundaries and significant semantic overlap, including infiltration and pneumonia. This limitation primarily arises from the reliance on image-level supervision, which lacks explicit lesion localization and anatomical priors. Future research could integrate weakly supervised lesion cues, anatomical structure guidance, and more interpretable visualization techniques to enhance the modeling of complex pathological patterns. All experiments were conducted at a standardized resolution of 224×224 , where CR-MSNet exhibited improved fine-grained feature modeling, particularly for small-lesion categories. However, employing higher resolutions or multi-scale strategies may further enhance sensitivity to subtle lesions and warrants systematic evaluation in future studies. Although validated on 2D CXR images, the dual-branch multi-scale and cross-attention framework of CR-MSNet is designed to be modality-agnostic and can theoretically be extended to 3D imaging modalities, such as CT or MRI, with appropriate architectural adaptations. Its scalability is promising; however, the effectiveness across modalities requires further empirical validation.

Conclusions

This study proposes CR-MSNet, an advanced framework designed for the classification of multi-label CXR images. The framework incorporates a dual-branch multi-scale attention network structure along with a PMS-CSA module. The dual-branch network facilitates unified modeling of multi-scale features, capturing both local detail and broader global contextual semantics, thereby accommodating the complex and diverse distribution of lesions in CXRs. Concurrently, the PMS-CSA module enhances feature representation through parallel multi-scale channel attention and spatial attention mechanisms, while integrating a cross-attention strategy to achieve adaptive fusion of features between branches. This approach further improves the network's overall discriminative

capability and prediction stability. Additionally, CR-MSNet utilizes a two-stage training strategy, effectively addressing the prevalent class imbalance issue in the ChestX-ray14 dataset. CR-MSNet presents a promising solution to critical challenges in multi-label CXR classification, including the unified modeling of multi-scale lesions, insufficient collaboration between local and global semantics, and model instability arising from class imbalance. The synergistic effects of these methodologies yield an average AUC of 0.847 on the ChestX-ray14 dataset, thereby validating its effectiveness and potential for application in automated multi-label CXR analysis. Future work will focus on further enhancing the model for deployment in real-time clinical settings.

Data availability

The datasets analyzed during the current study are available at the following links: <https://www.kaggle.com/datasets/nih-chest-xrays/data>

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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